



Multi-objective human-robot collaborative disassembly line balancing considering components remanufacture demand and hazard characteristics

Lixia Zhu^{a,b}, Yarong Chen^a, Jabir Mumtaz^{a,*}

^a College of Mechanical and Electrical Engineering, Wenzhou University, Wenzhou, China

^b Wenzhou Key Laboratory of Dynamics and Intelligent Diagnosis-Maintenance of Advanced Equipment, Wenzhou 325035, China

ARTICLE INFO

Keywords:

Human-robot collaboration
Multi-objective optimization
Disassembly line balancing
Remanufacture demand
Hazard characteristics
Teaching and learning optimization

ABSTRACT

Disassembly enterprises face significant challenges in managing increasingly intensive workloads with a workforce alone. To address this, current study focuses on balancing a human-robot collaborative disassembly line. For the task operator allocation problem, a mapping constraint mechanism between the disassembly tasks and the task operators is established based on the remanufacture demand and hazard characteristics of components. A mixed-integer programming model is formulated, encompassing four optimization objectives: the number of workstations, the task operator idle balancing index, the demand index, and the hazard index. A novel modified teaching and learning optimization approach is proposed for the addressed human-robot collaborative disassembly line balancing problem. A two-layer encoding strategy is designed based on the solving characteristics of the problem and an embedded perturbation strategy based on randomly transforming the task operator is introduced to enhance the optimization performance of the proposed method. The validity of the mixed-integer programming model and the efficacy of the proposed algorithm are demonstrated through two small-scale human-robot collaborative disassembly case studies. The proposed algorithm is then applied to two real-life cases of human-robot collaborative disassembly lines with different scales. The results of the proposed optimization method are compared with other advanced algorithms, demonstrating its superior performance in both single-objective and multi-objective optimization based on multiple evaluation indicators.

1. Introduction

The swift evolution of intelligent technologies and consumer behaviors has led to the significant development of electro-mechanical waste products. Typically, these discarded items encompass components with robust functionalities and recyclable materials, yet they harbor hazardous substances capable of environmental contamination. Suppose these huge waste products are directly discarded, incinerated, or landfilled. In that case, it is bound to cause serious harm to the ecological environment and a tremendous waste of reusable resources contained in these waste products. With the continuous strengthening of resource and environmental constraints, developing a green, low-carbon circular economy has become the fundamental way to break the resource and environmental constraints, cope with climate change and cultivate new economic growth points. The scarcity of resources and ecological degradation are pivotal constraints impeding the sustainable progression of society in the contemporary era. Hence, the timely and effective management of waste products stands as an urgent imperative

across nations (Du, Xu, & Li, 2018; Yin, Zhang, Zhang, Wu, & Liang, 2022). With the promulgation of relevant policies, regulations, and rules in various countries, the recycling industry for waste products has grown continuously, providing a platform for the standardized recycling of waste products (Hjorth & Chrysostomou, 2022). Disassembly is a crucial step for recycling waste products and a central link connecting the forward and the reverse logistics of products under the background of circular economy development (Bentaha, Dolgui, Battaia, Riggs, & Hu, 2018).

With the increasing number of waste products, the demand for disassembly continues to expand, putting forward requirements for large-scale and efficient disassembly production of waste products. As a more efficient disassembly production organization form, the disassembly line has gradually replaced the single disassembly workstation and the disassembly cell and has been popularized in disassembly enterprises. At present, the disassembly of waste products mainly depends on human workers. However, with the explosive growth of scrap products, coping with the increasingly intensive disassembly workloads that

* Corresponding author.

E-mail addresses: m18200289139@163.com (L. Zhu), yarongchen@126.com (Y. Chen), jabirmumtaz@live.com (J. Mumtaz).

<https://doi.org/10.1016/j.cie.2024.110621>

Received 4 June 2024; Received in revised form 13 September 2024; Accepted 30 September 2024

Available online 9 October 2024

0360-8352/© 2024 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

rely only on human workers is becoming difficult (Wu, Zhang, Yin, & Zhang, 2022). The increasingly intensive disassembly workloads aggravate the fatigue of the disassembly workers, resulting in a sharp decline in disassembly production efficiency (Lee, Behdad, Liang, & Zheng, 2022). In addition, some components are potentially harmful to human health, and some disassembly operations that do not conform to ergonomics threaten human workers' physical and mental health (Bentaha et al., 2018; Zeng, Zhang, Liang, & Zhang, 2023). The development of collaborative robots drives the rise of the human-robot collaborative (HRC) disassembly mode, which makes up for the shortcomings of the manual and robotic disassembly modes. The HRC disassembly mode combines the advantages of human workers' judgment and thinking ability and the safety, accuracy, and high efficiency of robots (Baratta, Cimino, Longo, & Nicoletti, 2023; Rinaldi, Caterino, & Fera, 2023; Yao, Li, Ji, Xiao, & Xu, 2024), and giving way to more significant disassembly production potential (Guo, Zhang et al., 2023; Wu, Zhang, Zeng, & Zhang, 2024).

Two types of task operators include human workers and robots on the HRC disassembly line, diverging from manual and robotic disassembly lines. These two task operators have their task operation capabilities and applicable characteristics. For instance, disassembly robots excel in handling tasks characterized by heavy loads, high repetition rates, and hazardous disassembly activities. Conversely, human workers demonstrate proficiency in intricate and delicate disassembly tasks, such as extracting rusty fasteners or removing deformed and aged components that might pose challenges for standard disassembly procedures. Variances in disassembly methods, time allocation, tool utilization, and other factors might arise when assigning the same disassembly task for a specific component to different task operators. These discrepancies can significantly influence the planning of disassembly sequences and the allocation of workstations for disassembly tasks. This variability can impact workload balancing among task operators on the HRC disassembly line and the overall efficiency of the HRC disassembly line. Hence, the challenge presented by the HRC disassembly line balancing problem (HRC-DLBP), which integrates multiple optimizing levels and dimensions, including task operator assignment, disassembly task sequence planning, and task allocation to workstations into a comprehensive decision-making framework, increases complexities.

On the other hand, various components of waste products have different hazard and demand characteristics, which are reflected in the fact that some components may produce substances that are harmful to human health or the ecological environment during the disassembly process. In contrast, some components can directly enter the remanufacture production process according to their end-of-life status. Therefore, from an economic and environmental perspective, the disassembly tasks corresponding to the hazardous components need to be assigned to robots as much as possible to reduce the harm to human workers' health. Besides, the disassembly tasks corresponding to the components with remanufacture demand characteristics need to be assigned to human workers for disassembly as much as possible to ensure that they can be removed intact and finely to create value for disassembly enterprises. Based on this consideration, this paper addresses the HRC-DLBP by leveraging the advantages inherent in the HRC disassembly production mode. Specifically, it accounts for the characteristics associated with human and environmental hazards and the demands for reusing and remanufacturing diverse components within waste products. A mapping constraint mechanism establishes a mapping correlation between the disassembly tasks with different characteristics and the two types of task operators to address the task operator assignment challenge for various components with distinct characteristics. A mixed-integer programming (MIP) model is established for the addressed HRC-DLBP. The MIP model incorporates minimizing the number of workstations, the task operator idle balancing index, the demand index, and the hazard index simultaneously. Therefore, the contributions of this research can be listed as follows:

- The HRC-DLBP is addressed, considering the task operator assignment issue for the disassembly tasks corresponding to the components with hazard and remanufacture characteristics.
- A mapping constraint mechanism is established between the disassembly tasks with different characteristics and the two types of task operators.
- An MIP model for HRC-DLBP is built considering the number of workstations, the task operator idle balancing index, the hazardous index and the demand index.
- A multi-objective modified teaching and learning optimization algorithm (MTLO) is proposed with an embedded perturbation strategy based on randomly transforming the task operator.

The remainder of the paper is presented in the following sequence. A review of relevant literature review is presented in Section 2. The problem formulation is elaborated and the MIP model for the addressed HRC-DLBP is established in Section 3. Section 4 expounds on the proposed MTLO algorithm. In Section 5, the correctness of the MIP model and the effectiveness of the proposed MTLO algorithm are verified through two small-scale HRC disassembly cases. Moreover, the superiority of the proposed MTLO algorithm in terms of solution performance based on multiple evaluation indicators is further confirmed by solving three case studies in this section. Afterwards, the summary of the presented research and future research directions are presented in the conclusion section.

2. Literature review

With the development of the circular economy, the disassembly line balancing problem, i.e., DLBP, has become a hot topic of discussion in academic research, attracting more scholars to research it. The existing literature research concerning DLBP can be divided into three categories based on the task operators: the manual DLBP, the robotic DLBP, and the human-robot collaborative DLBP. A classified overview of the relevant literature research on these three groups of DLBP is conducted in the following sections.

2.1. Manual DLBP

As far as the manual DLBPs are concerned, Qin et al. (2024) studied the disassembly line balancing problem considering the skill level and learning effect of the disassembly workers, and a mixed-integer programming model is established for this problem taking the disassembly profit as the optimization objective. To address this issue, an improved fruit fly optimization algorithm is proposed, and its effectiveness and efficiency are verified through comparative experiments with CPLEX and some popular algorithms. Zhou and Bian (2022a) presented a resource-constrained mixed-model multi-manned DLBP and established an MIP model based on the cycle time and the total number of workers and applied the epsilon constraint method to obtain the exact solutions to small-scale problems. An adaptive salp-swarm approach is proposed to solve large-scale problems, and the proposed algorithm's effectiveness is demonstrated using different evaluation metrics. Liang, Zhang, Zhang, Xu, and Yin (2023) addressed the multi-objective partial DLBP incorporating four optimization objectives, i.e., the workstation number, workload, the number of tool switches and the energy consumption. An MIP model and a multi-objective social spider algorithm are presented to optimize two typical benchmark instances with different scales and a partial disassembly instance of the printer. Çil, Kizilay, Li, and Öztop (2022) presented a two-sided DLBP considering the sequence-dependent setup time and established two new MILP models and a constrained programming model. An improved version of the genetic algorithm (GA) and artificial bee colony (ABC) algorithm is presented to solve the large-scale problem instances.

Moreover, Guo et al. (2023) tackled a stochastic DLBP with task failure risks to maximize the expected recovering profit. A hybrid meta-

heuristic method combining the variable neighborhood descent algorithm with the ABC algorithm is proposed for the developed model. Yin, Zhang, Wu, et al. (2023) addressed a multi-manned partial DLBP of type II to minimize the cycle time, the idle balancing index and the disassembly profit. An enhanced version of NSGA-II has been developed to address large-scale problems effectively. Besides, Guo et al. (2022) addressed the U-shaped DLBP to optimize the balance index of workers' fatigue and maximize the disassembly profits. An efficient multi-objective evolutionary approach with a collaborative resource allocation strategy is proposed to solve the problem. Wang et al. (2023) proposed the DLBP considering partial destructive disassembly mode and constructed a MILP model with the objectives of minimizing the number of workstations, the smoothness index, energy consumption and the disassembly benefits. An improved GA for solving large-scale problems was developed.

2.2. Robotic DLBP

Robotic disassembly line leverages robotic precision and consistency to handle complex, repetitive, and hazardous tasks, thereby enhancing safety and productivity in industrial settings. Regarding the balancing issue of the robotic disassembly line, Guo, Li, Du, Sun, and Wang (2024) focused on a multi-robot partial disassembly line balancing problem considering preventive maintenance scenarios taking into account the impact of machine failures caused by the inevitable aging of machines, in which four optimization objectives are considered: cycle time, disassembly profit, workstation reconfiguration time and cost. A multi-population cooperative coevolution artificial bee colony algorithm is proposed to solve the problem. Hartono, Ramírez, and Pham (2023) presented a multi-objective decision-making method based on the framework of robot disassembly units to seek to improve the economic benefits of disassembly, reduce energy consumption during the disassembly process, and alleviate the negative impact of disassembly production on the environment considering the robot disassembly sequence, component disassembly direction, component recycling method and the component disassembly tool multiple optimization aspects. An improved discrete bee algorithm was applied to solve the optimal robot disassembly planning scheme. Zhou and Bian (2022b) presented a sustainable robotic DLBP considering the uncertain processing time, the sequence-based tool changeover and the robots with different efficiencies and energy consumption rates considering the cycle time and the total energy consumption two optimization goals. The epsilon constraint algorithm helped to obtain the exact solutions for small-scale cases. A modified bi-objective Harris Hawks optimization algorithm is developed to solve large-scale instances.

Similarly, Zeng, Zhang, Yin, and Zheng (2022) focused on the robot disassembly line balancing and sequencing joint optimization problem considering the disassembly completion time, the energy consumption and the disassembly profit multiple objectives. A hybrid genetic simulated annealing method is presented to solve this complex problem. Fang, Ming, Li, Liu, and Pham (2020) aimed to research the multi-robot parallel DLBP considering the uncertain disassembly production environments. The multi-objective mathematical model for the addressed problem using uncertain disassembly task time is built to minimize the cycle time and energy consumption. A Pareto-based evolutionary algorithm hybridized with the simulated annealing algorithm is presented to solve the problem. Liu et al. (2020) studied the robotic disassembly line balancing and sequence planning joint optimization problem considering cycle time, number of robotic workstations and smoothness index as optimization objectives. A refined discrete bee algorithm incorporating variable neighborhood search is devised to address the collaborative optimization problem.

2.3. Human-robot collaborative DLBP

The HRC disassembly line leverages the strengths of humans and robots, with humans handling complex, adaptive tasks and robots performing repetitive or hazardous operations, resulting in a well-coordinated and balanced workflow (Kheirabadi, Keivanpour, Chiniyah, & Frayret, 2023). Guo et al. (2024) addressed the hybrid partial destructive HRC-DLBP, considering the non-disassemblability of components. A mixed-integer programming model is established to obtain the accurate number of workstations, the smoothness index, cost, and the negative impact of noise pollution on workers. An improved grey wolf optimization algorithm is proposed, and its superiority is verified through different cases and compared with multiple algorithms. Qin et al. (2024) studied the HRC-DLBP within a combined U-shaped and single straight-line layout. The authors developed a mixed-integer linear programming model to maximize disassembly profit by balancing the workload among workstations. Comparative experimental results of disassembly examples with different scales demonstrated that hybrid disassembly lines, comprising human and robot collaboration, have significant advantages over disassembly lines composed solely of human workers or robots when dismantling products. Jiao, Feng, and Yuan (2024) developed a model for balancing HRC disassembly lines considering load factors associated with human involvement such as the time-varying human conditions such as safety, cognitive behavior, workload, and human posture changes. The optimization objectives are to minimizing the cycle time, reducing the costs and the comprehensive smoothness index. An improved multi-objective fruit fly optimization algorithm is utilized to solve a case study involving the disassembly of a Mercedes battery pack, demonstrating the efficiency and robustness of the proposed algorithm.

Similarly, Zhang et al. (2024) studied a multi-product HRC-DLBP in a parallel layout to maximize disassembly profit and proposed an improved moth-flame optimization algorithm. The effectiveness of the proposed algorithm and its adaptability to the studied problem is proved by comparing it with other algorithms through multiple cases. Yin, Zhang, Liang, and Zeng (2023) addressed a multi-man-robot disassembly line balancing problem with the optimization objectives of minimizing the number of workstations, idle balance indicators for operators and the number of operators. A group evolutionary algorithm is proposed to check the correctness of the mathematical model and the performance of the proposed algorithm. Guo, Zhang et al. (2023) considered an HRC-DLBP model with stochastic task times to maximize total profit and minimize energy consumption. An improved Pareto-based multi-objective shuffled frog-leaping algorithm is introduced and validated by comparing with several popular and advanced methods through disassembly experiments on products of different sizes. Wu, Zhang, Zhang, and Zeng (2023) introduced human-robot interaction to two-sided disassembly lines and proposed a two-sided HRC-DLBP with human-robot interaction constraints. A mixed-integer programming model is established considering the number of workstations, smoothness index, robot energy consumption, and disassembly cost and an improved discrete squirrel search algorithm is developed. The results indicate that the proposed method can fully utilize the significant advantages of human-computer interaction technology in disassembly and assembly lines, especially in reducing energy consumption and cost expenditures. Wu et al. (2024) constructed a mixed-integer programming model based on the number of workstations, smoothness index, and cost indicators to address the balance problem of human-machine collaborative disassembly lines under human-computer interaction constraints. A hybrid local search genetic algorithm is developed, and the accuracy of the proposed MIP model and algorithm is verified through two HRC-DLBP examples. Xu et al. (2021) studied HRC-DLBP, focusing on safety and proposed an improved discrete bee algorithm to optimize the number of workstations, smoothness index, and demand

Table 1

A summary of HRC-DLBP related literature.

Reference	Task classification				Optimization objectives					Solving approaches	
	Hazardous task	Complex task	Demand task	Other tasks	F_1	F_2	F_3	F_4	Others	Mathematical programming method	Intelligent algorithm
(L. Guo et al., 2024)	✓	✓	—	Noise pollution	✓	—	—	—	Smoothness index, Costs, Noise pollution	MIP	Improved grey wolf optimization algorithm
(Qin, Xiang, et al., 2024)	—	—	—	—	—	—	—	—	Disassembly profit	MIP	—
(Jiao et al., 2024)	—	—	—	—	—	—	—	—	Cycle time, Associated costs, Smoothing exponent	—	Fruit fly optimization algorithm
(Zhang et al., 2024)	✓	✓	—	—	—	—	—	—	Disassembly profit	MIP	Moth-Flame algorithm
(Yin, Zhang, Liang, et al., 2023)	✓	✓	—	—	✓	✓	—	—	Number of operators	MIP	Group evolutionary algorithm
(Guo et al., 2023)	—	—	—	—	—	—	—	—	Total profit, Energy consumption	—	Shuffled frog leaping algorithm
(Wu et al., 2023)	✓	✓	—	—	✓	—	—	—	Smoothness index, Energy consumption, Cost	MIP	Discrete squirrel search algorithm
(Wu, Zhang, Zeng, & Zhang, 2024)	✓	✓	—	—	✓	—	—	—	Smoothness index, Various costs	MIP	Hybrid local search genetic algorithm
(Jin, Zhang, Fang, & Pham, 2022)	—	—	—	—	—	—	—	—	Cycle time, Total number of robots and humans,	—	Evolutionary dynamic optimization algorithm
(Xu et al., 2021)	—	—	—	—	✓	—	—	✓	Smoothness index	—	Improved discrete Bees algorithm
(Fang, Li, Wang, & Lu, 2024)	—	—	—	—	—	—	—	—	Cycle time, Robot energy consumption, Number of humans	MIP	Multi-fidelity optimization algorithm
(Wu et al., 2022)	✓	✓	—	—	✓	—	—	—	Smoothness of load, Cost of disassembly	—	NSGA-II
(Hu et al., 2024)	—	—	—	—	—	—	—	—	Disassembly time	—	An ontology and rule-based method
(Lou et al., 2024)	—	—	—	—	—	—	—	—	Benefit, Efficiency	—	Hybrid ant lion optimizer
(Qin, Li, et al., 2024)	—	—	—	—	—	—	—	—	Disassembly profit, Completion time	MIP	Gray wolf optimization algorithm
(Lee et al., 2022)	—	—	—	—	—	—	—	—	Disassembly time	MILP	—
(Parsa & Saadat, 2021)	—	—	—	—	—	—	—	—	Disassembly time	—	Modified GA
(Xu et al., 2020)	—	—	—	—	—	—	—	—	Disassembly time, cost, difficulty	—	Modified discrete bees algorithm
Current study	✓	—	✓	—	✓	✓	✓	✓	—	MIP	Modified teaching and learning optimization algorithm

F_1 = Number of workstations; F_2 = Idle balancing index of task operator; F_3 = Hazard index; F_4 = Demand index.

index. Numerical analysis on bearing couplers showed that the proposed algorithm outperformed others in running time, average inter-generational distance, and Pareto front quality.

Moreover, Hu et al. (2024) used a rule-based reasoning approach to investigate a general ontology model for HRC disassembly. Comparative analysis with representative optimization algorithms validated the method's feasibility and superiority in generating optimal disassembly plans. Lou, Tan, Zhang, Zhou, and Lv (2024) proposed a personalized disassembly sequence planning method for human-robot hybrid disassembly units considering ergonomics. They established a bi-objective model to optimize profit and efficiency, addressing sequence scheduling and task allocation. A hybrid multi-objective ant-lion optimizer is introduced to maximize both objectives. Qin et al. (2024) studied HRC disassembly sequence planning considering component characteristics and established a mathematical model to maximize profit and minimize work time. The grey wolf optimization algorithm is employed and tested on various scales and scenarios. Parsa and Saadat (2021) addressed the disassembly sequence planning problem under the HRC disassembly mode. They evaluated the re-manufacturability of components based on

cleanability, reparability, and economy. A modified GA based on the PROMETHEE method is proposed to optimize disassembly time and total cost. Xu et al. (2020) presented the HRC disassembly sequence planning problem considering the difficulty of disassembly tasks. The HRC disassembly sequence planning model is developed considering the disassembly time, cost and difficulty. An improved Pareto-based discrete bee algorithm is proposed to optimize the quality of the solution. Due to the growing importance of recent industrial trends towards HRC-DLBP, we have focused on this issue in the current study. To address real-life industrial challenges effectively, we have selected four major objectives: the number of workstations, idle balancing index of task operator, hazard index, and demand index. Additionally, the choice of an optimization approach is crucial, and our selected method is discussed below.

2.4. Optimization approaches

The complexity of the HRC-DLBPs solving not only originates from the NP-hard characteristic of discrete combinatorial optimization

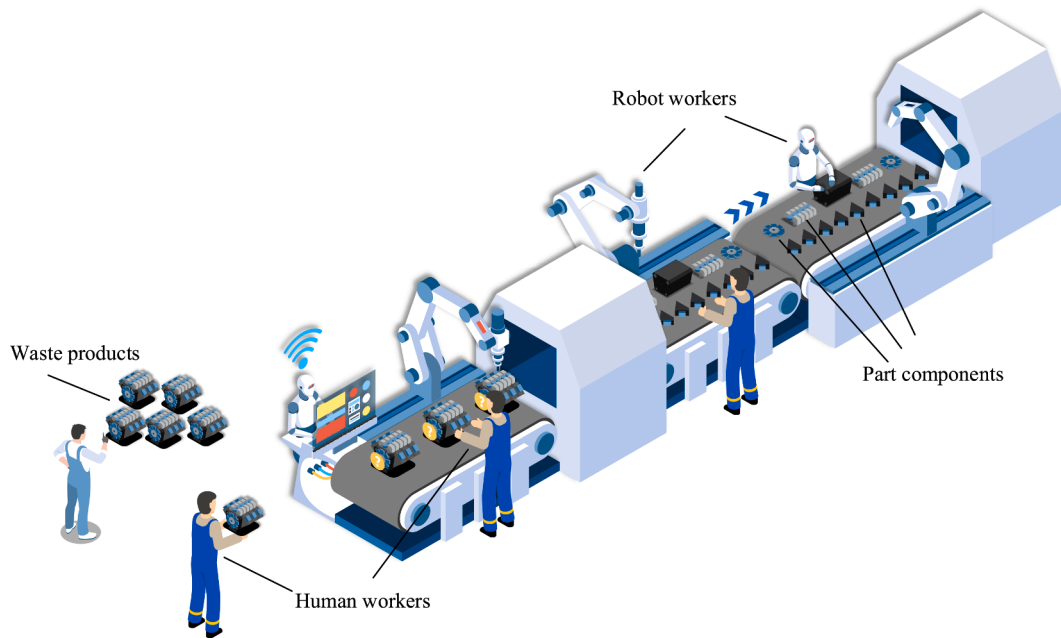


Fig. 1. The schematic diagram of the human-robot collaborative disassembly line.

problems (Liu et al., 2024) which means the complexity of the problem increases exponentially as the problem size expands, but also from the multi-level coupling solving characteristic of the HRC-DLBPs which means the HRC-DLBPs solving is more complex than the general DLBPs. The various approaches offer different strengths and can be selected based on the specific requirements of the HRC-DLBPs. As the field continues to evolve, integrating advanced optimization techniques and real-time adaptive methods will further enhance the capabilities of HRC-DLBPs solving. Mathematical programming models, such as MIP (Liu et al., 2024; Qin, Xiang, et al., 2024) and MILP (Lee et al., 2022), have been widely used to formulate HRC-DLBPs. These models aim to find optimal solutions for the HRC-DLBPs by defining a set of constraints and objective functions. While effective for small-scale problems, their computational complexity often limits their applicability to larger and more complex scenarios.

Metaheuristic algorithms are frequently employed to overcome the limitations of mathematical programming methods. These algorithms, such as GA (Parsa & Saadat, 2021; Wu, Zhang, Zeng, & Zhang, 2024) neighborhood search algorithm (Zhou, Du, Liu, & Shen, 2024), artificial bees colony algorithm (Mumtaz, Minhas, Rauf, Yue, & Chen, 2024; Xu et al., 2021), artificial fish swarm algorithm (Wang, Chen, Mumtaz, & Zhu, 2023), spider monkey optimization algorithm (Mumtaz et al., 2019; Mumtaz, Guan, Yue, Zhang, & He, 2020), evolutionary algorithm (EA) (Jin, Zhang, Fang, & Pham, 2022) and frog leaping algorithm (Guo, Li et al., 2023), are capable of finding the optimal solutions for small-scale problems and can provide optimal (near-optimal) solutions within reasonable time frames for large-scale problems. Besides, given the multiple objectives often involved in HRC-DLBPs, the metaheuristic algorithms integrated with the Pareto theory based multi-objective optimization method can efficiently optimize multiple objectives simultaneously, providing decision-makers with diverse solutions, which cannot be achieved by mathematical programming method. These algorithms are valuable for decision-makers considering various performance metrics simultaneously. Due to the advantages of the metaheuristic algorithms, a modified teaching and learning optimization algorithm, i.e., MTLO is proposed by integrating GA operations (Mumtaz et al., 2018), such as the crossover operation to deal with the complex multi-objective optimization of HRC-DLBP addressed in this paper.

2.5. Research gap

To date, an increasing number of studies have focused on HRC-DLB problems. Table 1 summarizes the latest literature studies highlighting the focus areas such as task classifications, optimization objective functions and different methodologies to solve the HRC-DLBPs. It can be seen from Table 1 that some studies do not consider the classification of tasks regarding the task operator allocation. Instead, the tasks are directly assigned to human workers or robots, while most studies classify tasks into complex and hazardous tasks. In addition, Wu et al. (2023) considered task operator allocation with human-robot interaction constraints. Xu et al. (2020) introduced disassembly parameters to quantify the difficulty of disassembly tasks and determine the task operator accordingly. Limited literature considers the allocation of task operators for the disassembly tasks corresponding to the components with remanufacture demand. Regarding optimization objectives, most research prioritizes the number of workstations, smoothness index, disassembly costs, and disassembly time. Few studies consider optimizing the disassembly priority for hazardous and remanufacturable components.

3. Problem formulation

The schematic diagram of the HRC disassembly line addressed in this paper is shown in Fig. 1. There are human workers and robots task operators on the HRC disassembly line, and each workstation on the HRC disassembly line can be equipped with up to one human worker and one robot to perform the disassembly tasks in parallel. To tackle the task operator allocation problem of the disassembly tasks for the HRC-DLBP, all disassembly tasks for the waste products are classified into three types based on the hazard and remanufacture demand characteristics of the components corresponding to the disassembly tasks: a) the disassembly tasks corresponding to the hazardous components are only allowed to be performed by the robots; b) the disassembly tasks corresponding to the components with remanufacture demand are only allowed to be performed by the human workers; c) the other disassembly tasks that the robots or the human workers can perform. Considering the task execution abilities of the human workers and the robots, a mapping constraint mechanism is established by reflecting the matching relationships between the task operators and the disassembly tasks with

different characteristic attributes. On this basis, in combination with the deduction of the disassembly precedence relationships among the tasks of the waste products, the disassembly tasks are allocated to each workstation of the HRC disassembly line under the constraint of cycle time, aiming to seek the HRC disassembly line balancing schemes that minimize the total workstations, the task operator idle balancing index, the hazard index and the demand index. The following assumptions are made to simplify the complex situation in solving HRC-DLBP.

- The complete disassembly mode is considered, i.e., all the components are removed, and the disassembly task corresponding to each component can only be assigned to one task operator for independent execution, i.e., there are no tasks involving human-robot interaction.
- The task time for workers and robots to remove each component is known.
- The task operator's movement time and tool replacement time are included in the task time.
- The physical implementation details in the disassembly process ensure effective collaboration between human workers and robots.
- The uncertainties and emergencies during the HRC disassembly process are ignored.

The symbols, parameters and decision variables required to formulate the MIP model for the HRC-DLBP are listed as follows:

Indices	
i, j	Indices for the disassembly tasks, $i, j \in \{1, 2, \dots, I\}$
p	Indices for the sequence number, $p \in \{1, 2, \dots, I\}$
k	Indices for the HRC disassembly workstations, $k \in \{1, 2, \dots, K\}$
w	Task operator, if the task operator is the robot, $w = 1$, and if the task operator is the human worker, $w = 2$
Parameters	
CT	Cycle time of line
t_i^w	The operation time required for the task i to be executed by the task operator w
D_i	Demand indicator, if the component corresponding to the disassembly task i is remanufacturable, $D_i = 1$; otherwise, $D_i = 0$
H_i	Hazard indicator, if the component corresponding to the disassembly task i is hazardous, $H_i = 1$; otherwise, $H_i = 0$
t_{bi}	Starting time of task i
M	A significant positive number
PD	The disassembly precedence relationships matrix $PD = [pd_{ij}]_{I \times I}$, if task i immediately precede the task j , $pd_{ij} = 1$; otherwise, $pd_{ij} = 0$
Decision variables	
Z_k	Binary variable, if the HRC disassembly workstation k is open, $Z_k = 1$; otherwise, $Z_k = 0$
U_k^w	Binary variable, if the workstation k is equipped with the task operator w , $U_k^w = 1$; otherwise, $U_k^w = 0$
x_{ik}^w	Binary variable, if the task i is assigned to the task operator w in the workstation k , $x_{ik}^w = 1$; otherwise, $x_{ik}^w = 0$
Q_{ijk}^w	Binary variable, if the disassembly tasks i and j are assigned to the same task operator w in the same workstation k and task i is operated earlier than task j , $Q_{ijk}^w = 1$; otherwise, $Q_{ijk}^w = 0$
s_{ip}	Binary variable, if the disassembly task i is assigned to the p^{th} sequence position, $s_{ip} = 1$; otherwise, $s_{ip} = 0$

Objective functions

$$F_1 = \sum_{k \in K} Z_k \quad (1)$$

$$F_2 = \sum_{k \in K} \sum_{w \in \{1, 2\}} \left(CT \cdot Z_k \cdot U_k^w - \sum_{i \in I} x_{ik}^w \cdot t_i^w \right)^2 \quad (2)$$

$$F_3 = \sum_{i \in I} \sum_{p \in I} s_{ip} \cdot H_i \cdot p \quad (3)$$

$$F_4 = \sum_{i \in I} \sum_{p \in I} s_{ip} \cdot D_i \cdot p \quad (4)$$

The Equations from (1) to (4) delineate the four optimization objectives respectively: reduce the total number of HRC disassembly workstations, the task operator idle balancing index, the demand index, and the hazard index. From the perspective of disassembly costs, it is necessary to reduce the number of workstations opened as much as possible to minimize the disassembly costs as shown in Equation (1). On the other hand, to reduce the number of task operators and make the workload of each operator as full and balanced as possible, the task operator idle balancing index should be minimized as much as possible, as indicated in Equation (2). In addition, considering that the removal of hazardous components may cause damage and contamination to the remaining components of waste products, the disassembly tasks corresponding to the hazardous components should be performed as early as possible, as shown in Equation (3). Equation (4) represents the components with remanufacture demand should be removed as early as possible. The model constraints are:

$$\sum_{k \in K} \sum_{w \in \{1, 2\}} x_{ik}^w = 1 \quad \forall i \quad (5)$$

$$\sum_{k \in K} \sum_{w \in \{1, 2\}} x_{ik}^w \cdot H_i \cdot w = 1 \quad \forall i \in \{i | H_i = 1\} \quad (6)$$

$$\sum_{k \in K} \sum_{w \in \{1, 2\}} x_{ik}^w \cdot D_i \cdot w = 1 \quad \forall i \in \{i | D_i = 1\} \quad (7)$$

$$Z_k \leq \sum_{w \in \{1, 2\}} U_k^w \leq 2 \cdot Z_k \quad \forall k \quad (8)$$

$$\sum_{i \in I} \sum_{w \in \{1, 2\}} x_{ik}^w \leq I \cdot U_k^w \quad \forall k \quad (9)$$

$$\sum_{i \in I} \sum_{w \in \{1, 2\}} x_{ik}^w \geq U_k^w \quad \forall k \quad (10)$$

$$\sum_{i \in I} \sum_{w \in \{1, 2\}} x_{ik}^w \leq I \cdot Z_k \quad \forall k \quad (11)$$

$$\sum_{i \in I} \sum_{w \in \{1, 2\}} x_{ik}^w \geq Z_k \quad \forall k \quad (12)$$

$$Z_{k-1} \geq Z_k \quad \forall k \in \{2, 3, \dots, K\} \quad (13)$$

$$t_{bj} \geq t_{bi} + \sum_{k \in K} \sum_{w \in \{1, 2\}} t_i^w \cdot x_{ik}^w \quad \forall i, j \in \{I | (pd_{ij} = 1)\} \quad (14)$$

$$M(1 - Q_{ijk}^w) + t_{bi} + t_i^w \cdot x_{ik}^w \geq t_{bj} + t_j^w \cdot x_{jk}^w \quad \forall i, j, i \neq j, \forall k, \forall w \quad (15)$$

$$Q_{ijk}^w + Q_{jik}^w \leq \frac{1}{2} (x_{ik}^w + x_{jk}^w) \quad \forall k, \forall w, \forall i, j, i \neq j \quad (16)$$

$$Q_{ijk}^w + Q_{jik}^w + 1 \geq x_{ik}^w + x_{jk}^w \quad \forall k, \forall w, \forall i, j, i \neq j \quad (17)$$

$$t_{bj} \geq CT \cdot \left(\sum_{k \in K} \sum_{w \in \{1, 2\}} x_{jk}^w \cdot k - 1 \right) + \sum_{k \in K} \sum_{w \in \{1, 2\}} \sum_{i \in I, i \neq j} (Q_{ijk}^w \cdot t_i^w) \quad \forall j \quad (18)$$

$$t_{bi} + \sum_{k \in K} \sum_{w \in \{1, 2\}} x_{ik}^w \cdot t_i^w \leq CT \cdot \sum_{k \in K} \sum_{w \in \{1, 2\}} x_{ik}^w \cdot k \quad \forall i \quad (19)$$

$$t_{bi} \geq 0 \quad \forall i \quad (20)$$

As indicated in Equation (5), any disassembly task must be allocated to a specific task operator at a workstation. Equation (6) suggests that the disassembly tasks corresponding to the hazardous components must be assigned to the robot. Equation (7) ensures that the disassembly tasks corresponding to the components with remanufacture demand must be assigned to the human workers. Equation (8) indicates that if a workstation is open, it must be equipped with at least one type of task operator and can be equipped with a maximum of two task operators of

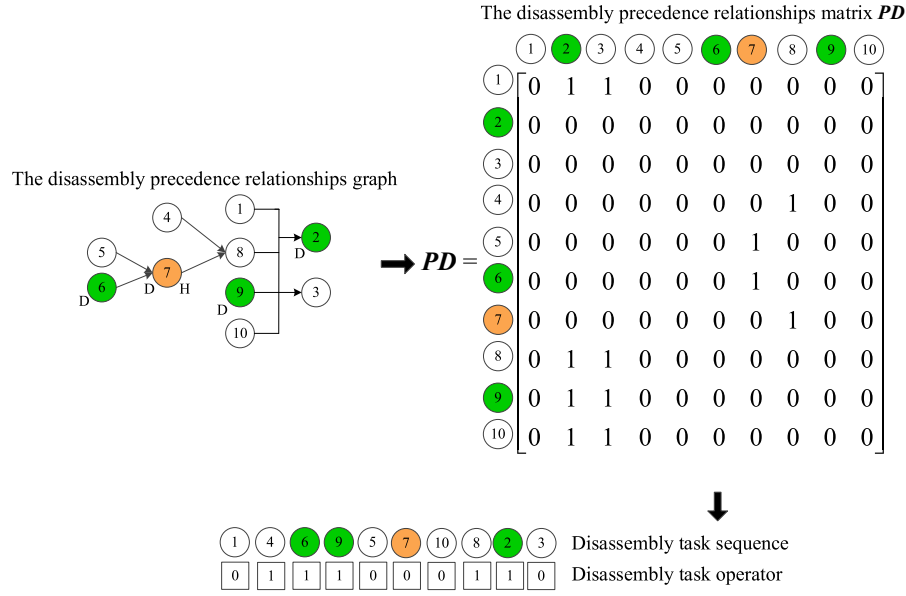


Fig. 2. The encoding of the solution for the HRC-DLBP.

different types. Equations (9)-(10) indicate that if a workstation is equipped with one type of task operator, there must be at least one piece of disassembly task assigned to the task operator, and the total number of tasks assigned to the task operator cannot surpass the overall count of disassembly tasks. Equations (11)-(12) indicate that if a human-machine collaborative disassembly workstation is open, there must be at least one piece of disassembly task allocated to the workstation, and the total number of tasks assigned to the workstation cannot surpass the overall count of disassembly tasks. Equation (13) implies no empty workstation exists as workstations are open in sequence.

Equation (14) details the sequential order of operations for tasks i and j , where task i immediately precedes task j in disassembly. Specifically, the start time of the task j must not be earlier than the end time of the execution disassembly task i . Equation (15) represents that if any two different disassembly tasks i and j are assigned to the same task operator of a workstation and task i is allocated to the task operator earlier than task j , task j cannot be performed until task i is completed; Equations (16)-(17) indicate that if the two disassembly tasks i and j are allocated to the same task operator of a workstation, these two disassembly tasks i and j cannot be operated simultaneously in parallel. Equation (18) denotes that if the two different disassembly tasks i and j are allocated to the same task operator of a workstation and task i is

allocated to the task operator earlier than task j , the start time of task j must be greater than the sum of the startup time of the corresponding workstation and the disassembly time of task i . Equation (19) states that any disassembly task i must be completed within the HRC disassembly line cycle time. Equation (20) indicates that the start time of any task i is a positive real number.

4. The proposed MTLO algorithm

The HRC-DLBP exhibited discrete combinatorial and multi-objective optimization traits. The MTLO algorithm, as proposed, incorporates a two-layer encoding strategy tailored to the problem's solving characteristics. The two-layer encoding strategy encapsulates task operators' assignment to disassembly tasks and the parallel disassembly sequence, comprehensively representing the problem's intricacies. The integration of two-point crossover and single-point insertion operations are introduced during the teaching and mutual learning phases of the proposed algorithm. These operations aim to enhance the algorithm's global optimization capabilities and expedite its convergence rate (Zhu, Zhang, & Guan, 2020). The local search strategy based on the two-point swap operation is applied in the self-learning stage. The perturbation strategy based on randomly transforming the task operator is integrated into the

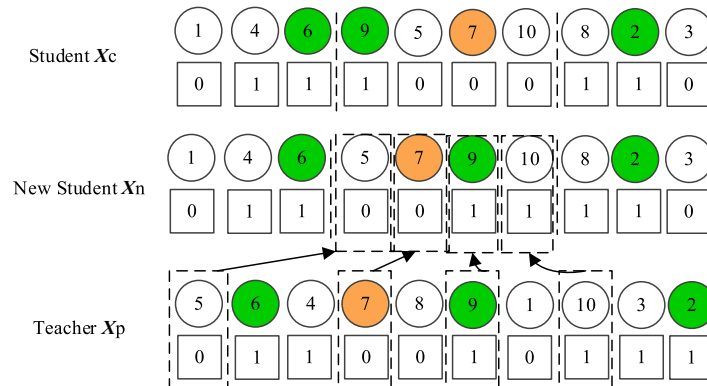


Fig. 3. The two-point crossover operation.

three stages of teaching, mutual learning, and self-learning to effectively enhance the local optimization performance of the presented MTLO algorithm. The following is a specific explanation of the proposed MTLO algorithm.

4.1. Encoding and initialization

Considering the multi-level coupling optimization characteristic, a two-layer encoding method is innovated with the dual information of the task operator assignment and the HRC parallel disassembly

with remanufacture demand, while those corresponding to task 7 are hazardous. In the second level of the encoding information, the number below each task represents the task operator allocated to the task, where 0 and 1 illustrate the robot and the human worker, respectively. The HRC parallel disassembly sequences are randomly generated while adhering to the constraints of disassembly precedence relationships. The procedure for generating the initial individuals is given below.

Algorithm 1: Initial population

Input: Total disassembly tasks I , precedence matrix PD , hazard index and demand index information of each disassembly task, total individuals p_num

For $i=1$ to p_num

For $j=1$ to I

 As per relation matrix PD , the tasks whose direct priority disassembly task set is not allocated are found out, and the task set CS to be assigned is formed;

 A random task T is selected in CS and assigned to the current sequence at position j in the current individual Pop_i ;

If parts/components of task T are hazardous

 Task T is allocated to the robot;

Else

If the parts/components of task T are reusable or re-manufacturable Task T is allocated to the human worker;

Else

 The task T is randomly assigned to a specific type of task operator;

End

End

 Renew the PD and CS ;

End For j

End For i

Output: The initial population Pop

sequence. Each individual of the MTLO algorithm corresponds to an HRC parallel disassembly sequence, which contains two levels of information: the task operator assignment for each task and the parallel task sequence. Within the initial level of the encoding scheme, each circle represents a disassembly task and the numerical value inside the circle represents the respective task number. In the disassembly precedence relationships graph, there are two letters on the left and right sides below each circle, where the letter D indicates that the component corresponding to the task is remanufacturable and the letter H indicates that the component corresponding to the task is hazardous. When a component possesses both remanufacture demand and hazard characteristics, the component will be treated as a hazardous component and the demand characteristic will be ignored. As depicted in Fig. 2, the components corresponding to the disassembly tasks 2, 6 and 9 are those

4.2. Decoding

It is significant to allocate the tasks of the HRC parallel disassembly sequence, matching each individual to specific workstations, so that the objective function value for each optimization objective of each individual, which means the knowledge level of each individual can be calculated and the Pareto comparison can be executed. Therein, the Pareto non-dominated individuals are the teacher individuals, while the others are the student individuals. The specific decoding process is given below.

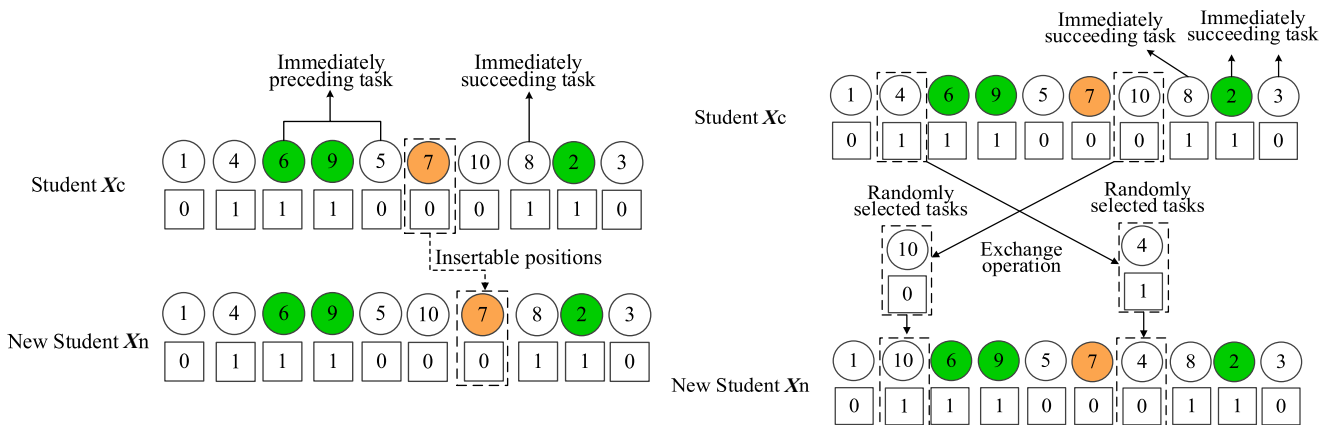


Fig. 4. Mutation operations (a) The single-point insertion (b) The two-point swap.

Algorithm 2: Decoding

Input: Feasible HRC task sequence (X), total tasks (I), operation time of each task, cycle time (CT)

Start the first HRC workstation, i.e., let $k = 1$;

Let the operation time of human worker at workstation k to 0, i.e., let $WT_k_H = 0$; thus the remaining operation time of human worker at k is equal to CT , i.e., let $RT_k_H = CT$; Let the total working time of robot worker at k to 0, i.e., let $WT_k_R = 0$, and then the remaining working time of the robot worker at k is equal to CT , i.e., let $RT_k_R = CT$;

For $j = 1$ to TS

Determine the task operator of task j ;

If the task operator of the task j is the human worker

Determine the workstation and the starting time of task j FT_k_H by determining the immediate precedence tasks of task j and the task allocation status of the corresponding workstation;

Compare the manual operation time t_j_H of the task j and the remaining operation time of the human worker at the workstation k RT_k_H ;

If $t_j_H > RT_k_H$

Start a new workstation, let $k = k + 1$, and perform the procedure shown in Step (2);

Else

If $t_j_H \leq RT_k_H$

The task j is allocated to the k ;

Renew the total operation time of human worker at k , i.e., let $WT_k_H = WT_k_H + t_j_H$, and meanwhile renew the remaining operation time of the human worker at the workstation k , i.e., let $RT_k_H = CT - WT_k_H$;

End If

End If

Else

Determine the workstation and the starting time of task j FT_k_R by determining the immediate precedence tasks of task j and the task allocation status of the corresponding workstation;

Compare the operation time t_j_R of the task j and the remaining working time of the robot worker of the current workstation k RT_k_R ;

If $t_j_R > RT_k_R$

Open a new workstation, set $k = k + 1$, and perform the procedure shown in Step (2);

Else

If $t_j_R \leq RT_k_R$

Assign task j to the current k ;

Update the total operation time of the robot worker of the current k , i.e., set $WT_k_R = WT_k_R + t_j_R$, and update the remaining working time of the robot worker of the current workstation k at the same time, i.e., set $RT_k_R = CT - WT_k_R$;

End If

End If

End For j

Output: The task allocation scheme of the feasible HRC task sequence X

4.3. Teaching stage

In the teaching stage of the proposed MTLO algorithm, the students (dominated individuals) learn from the teachers (non-dominated individuals) to improve themselves. The teachers continue to improve the knowledge level in each iteration, thus forming a virtuous circle through which the students' knowledge levels are continuously improved. The teaching stage is shown in Equation (21).

$$\begin{cases} \text{new student}_i = \text{student}_i + \text{diff} \\ \text{diff} = r_i(\text{teacher} - TF \times \text{mean}) \\ TF = \text{round}[1 + \text{rand}(0, 1)] \end{cases} \quad (21)$$

The new and existing students represent the individuals with improved knowledge levels in the teaching stage and those with original knowledge levels, respectively. The teachers indicate the Pareto non-dominated individuals filtered under the current iteration, r is the learning step, a random number in the interval (0, 1), TF indicates the teaching factor, and mean is the students' average knowledge level. Considering the solving characteristics of the addressed HRC-DLBP, the

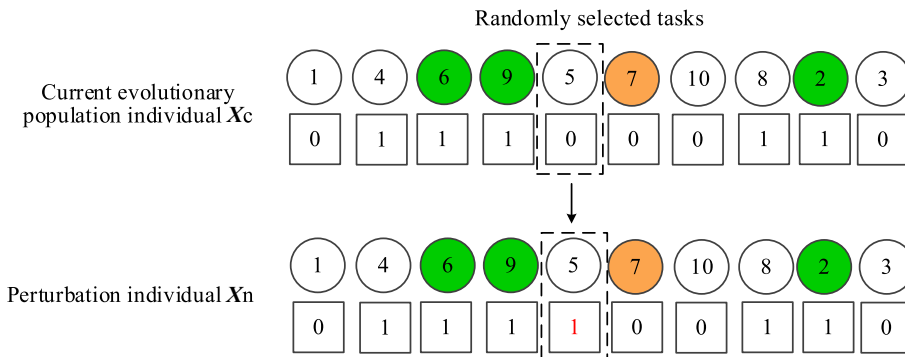


Fig. 5. The perturbation strategy based on random transformation of the task operators.

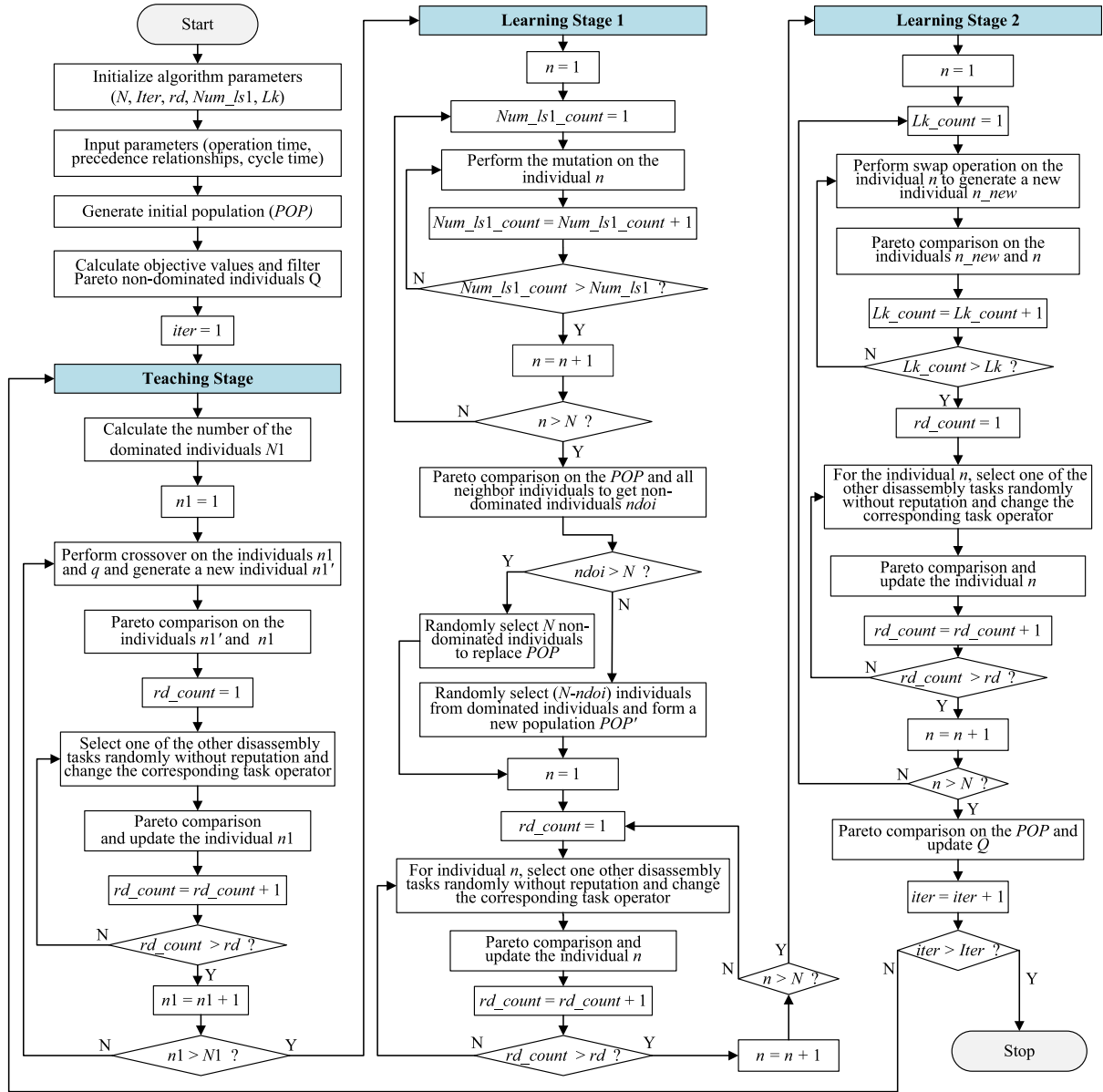


Fig. 6. The flowchart of the proposed MTLO algorithm.

two-point crossover operation is introduced in the teaching stage of the proposed MTLO algorithm to enable each student to interact with the teacher, through which the students can learn from the excellent teachers and improve their knowledge level effectively. The two-point

crossover operation is shown in Fig. 3. The specific procedures for the teaching stage of the MTLO algorithm are presented in the pseudo-code below.

Algorithm 3: Teaching stage of MTLO algorithm

Input: individuals pop (number of individuals is p_num), filtered Pareto non-dominated population individuals Q_pop (number of filtered Pareto non-dominated population individuals is Q_num)

For $i = 1$ to p_num

If the i^{th} population individual X_c is a student (The dominated individuals under the current iteration)

 Select a teacher X_p (the Pareto non-dominated population individual) from the set Q_pop and randomly choose two sequence positions c and d ($1 \leq c < d \leq TS$);

 Reorder the tasks and the task operators between the two sequence positions c and d in the student individual X_c according to the corresponding task order (together with the task operators) in the teacher individual X_p , thus generate the new student X_n ;

End If

End For i

Output: the new population individuals pop n

4.4. Learning stage

The learning stage includes two sub-stages: mutual and self-learning stages. In the mutual learning stage, the students learn from each other and constantly improve themselves. In the self-learning stage, the students review and consolidate their knowledge according to their situations, a meaningful way to enhance their knowledge level. According to the learning stage, combined with the characteristics of the addressed problem, a two-stage learning strategy is proposed to accelerate the global convergence performance of the proposed MTLO algorithm. In the first stage of the learning strategy, a greedy mechanism-driven local search operation based on the single-point insertion operation is presented, as shown in Fig. 4 (a). In the second stage, a local search strategy based on the two-point swap operation is adopted, as shown in Fig. 4 (b). The pseudo-code of the two-stage learning strategy is given below.

Algorithm 4: Two-stage learning strategy

Input: the individuals pop (number of individuals is p_num)

LEARNING STAGE 1

For $i = 1$ to p_num

For $j = 1$ to Num_ls1

 Select a sequence position randomly in the current population individual X_c ;
 Determine the immediate precedence disassembly task and the immediate succeeding disassembly task corresponding to the randomly selected sequence position;
 Ensure the insertable sequence positions of the disassembly task corresponding to the randomly selected sequence position;
 Insert the task into any one of the determined insertable sequence positions and generate the new student X_n ;
 Conduct the Pareto comparison on X_n and X_c ;
 If X_n Pareto dominates X_c
 X_c is replaced by X_n ;
 End If
End For j

End For i

LEARNING STAGE 2

For $i = 1$ to p_num

For $Lk_count = 1$ to Lk

 Randomly choose two disassembly tasks in X_c and find the immediate precedence and succeeding disassembly tasks of two selected disassembly tasks, respectively;
 If the exchange of these two selected disassembly tasks does not violate the disassembly precedence;
 Perform the two-point swap operation on X_c and generate the new population individual X_n ;
 Else
 Randomly reselect two tasks in X_c and find the immediate precedence and succeeding disassembly tasks of the two selected tasks, respectively;
 End If
 Conduct the Pareto comparison on X_n and X_c ;
 If X_n Pareto dominates X_c
 X_c is replaced by X_n ;
 End If
End For Lk_count

End For i

Output: the new individuals pop'

4.5. Perturbation strategy

On the HRC line, except for the disassembly tasks corresponding to the hazardous components that must be assigned to the robot. The disassembly tasks corresponding to the components with remanufacture demand must be assigned to the human worker. The disassembly tasks corresponding to the other components are randomly assigned to the robot or the human worker. Different task operator assignment schemes affect the HRC disassembly line balancing scheme. Taking this into account, a perturbation strategy involving the random transformation of task operators is proposed. This strategy targets the disassembly tasks related to components other than the hazardous ones and those with remanufacture demand. Specifically, for each perturbation, one of the disassembly tasks associated with the other components is randomly selected (task 5 is selected as shown in Fig. 5) in the current HRC task sequence and the task operator for the selected task (the task operator corresponding to the task 5 is transformed from the robot to the human

Table 2

The related data for the personal computer HRC disassembly case.

Task	Time (sec)		D	H	Immediate precedence disassembly tasks
	Human	Robot			
1	14	11	1	0	—
2	10	8	1	0	1
3	12	10	0	0	1
4	18	14	0	0	7
5	23	18	0	0	1
6	16	13	1	0	2, 3
7	20	16	0	1	8
8	36	29	0	0	2, 5, 6

worker). The schematic diagram of the perturbation strategy based on random transformation of the task operators is shown in Fig. 5.

4.6. Implementation steps of the MTLO algorithm

The implementation of the MTLO algorithm is carried out as follows:

Step 1: Initialization of parameters such as the number of individuals N , number of iterations $Iter$, number of perturbations rd , parameters for learning stage 1 Num_ls1 and parameters for learning stage 2 Lk . Input parameters: the operation time (human worker/robot), the disassembly precedence relationships among tasks, and predetermined HRC disassembly line cycle time.

Step 2: Randomly generate the initial population of individuals according to the method described in Section 4.1. Calculate the objective function values of each population individual according to the method described in Section 4.2 and filter the Pareto non-dominated individuals.

Step 3: Set the iteration count $iter = 1$.

Step 4: In the teaching stage, for each student individual, execute the random neighborhood search operation based on the two-point crossover operation described in Section 4.3 and conduct the Pareto comparison on the current and newly generated student individuals. Renew the initial student individual using the filtered non-dominated student individual.

Table 3

The results obtained by LINGO and MTLO algorithm for the personal computer HRC disassembly case.

Solution	MTLO				Time(sec)	LINGO				Time(sec)
	F_1	F_2	F_3	F_4		F_1	F_2	F_3	F_4	
1	4	126	7	7	2.61	4	—	—	—	1
						—	126	—	—	145
						—	—	7	—	1
						—	—	—	7	1

Table 4

The related data for the theoretical case.

Task	Time (sec)		D	H	Immediate precedence disassembly tasks
	Human	Robot			
1	14	11	0	0	—
2	10	12	0	0	1, 8, 9, 10
3	12	10	0	0	1, 8, 9, 10
4	17	14	0	0	—
5	23	25	0	0	—
6	14	11	0	0	—
7	19	15	0	1	5, 6
8	36	29	0	0	4, 7
9	14	11	1	0	—
10	10	12	0	0	—

Table 5

The solution results obtained by Lingo software and the proposed MTLO algorithm.

Solution	MTLO				Time (sec)	LINGO				Time (sec)
	F_1	F_2	F_3	F_4		F_1	F_2	F_3	F_4	
1	4	298	4	1	8.1	2	—	—	—	3
2	2	46	5	2		—	9	—	—	947
3	3	687	4	1		—	—	3	—	1
4	2	46	4	5		—	—	—	1	1
5	3	570	4	2						
6	3	382	3	5						
7	2	9	4	6						
8	3	516	3	4						
9	3	516	4	3						
10	4	291	4	3						
11	4	291	3	4						
12	2	9	6	3						
13	3	45	5	2						
14	3	45	4	5						

Step 5: Implement rd times perturbation strategy as Section 4.5 without repetition on each individual generated in Step 4. Conduct the Pareto comparison and renew the initial individual using the filtered non-dominated individual.

Table 6

The related data for the CRT TV HRC disassembly case.

Task	Time (sec)		D	H	Immediate precedence disassembly tasks	Task	Time (sec)		D	H	Immediate precedence disassembly tasks
	Human	Robot					Human	Robot			
1	50	40	0	0	—	15	4	3	0	1	—
2	9	7	0	0	1	16	4	3	0	0	8, 9, 15
3	3	2	0	0	2	17	4	3	1	0	13
4	7	6	1	0	2	18	4	3	0	0	10
5	24	19	0	0	2	19	17	14	0	0	2
6	2	2	0	1	5, 7	20	16	13	0	0	19
7	10	8	0	0	2	21	7	6	0	1	10, 16, 19
8	18	14	1	0	15	22	3	2	0	1	21
9	4	3	1	0	15	23	3	2	0	1	22
10	4	3	0	0	5	24	4	3	0	0	22
11	4	3	1	0	10	25	2	2	0	0	22
12	4	3	0	0	10	26	2	2	1	0	22
13	4	3	1	1	10	27	2	2	0	0	23
14	4	3	1	1	10						

Step 6: In learning stage 1, execute the greedy mechanism-driven local search based on the single-point insertion operation described in Section 4.4 on each individual generated in Step 5 to generate Num_{ls1} neighbor individuals and conduct the Pareto comparison on the current and newly generated neighbor individuals. Renew all the individuals using the filtered non-dominated individuals. Perform the perturbation strategy as in Section 4.5 without repetition of each individual rd times and update the individual.

Step 7: In learning stage 2, perform the local search strategy based on the two-point swap operation as Section 4.4 on each individual generated in step 6 and continuously update the knowledge level of the individuals. On this basis, the perturbation strategy described in Section 4.5 is performed without repetition on each individual rd times and updates the individual.

Step 8: Update the current Pareto non-dominated individuals by performing a Pareto comparison on the current population individuals.

Step 9: $iter = iter + 1$, if $iter < Iter$, return to Step 4; else, follow Step 10.

Step 10: Obtain the final Pareto non-dominated individuals and terminate the algorithm iteration procedure.

To demonstrate the implementation steps of the algorithm more intuitively, Fig. 6 exhibits a flowchart of the proposed MTLO algorithm, in which the overall process of the algorithm is shown.

5. Computational experiments and case study

5.1. Model validation and algorithm effectiveness for small-scale cases

To verify the correctness of the MIP model established for the addressed HRC-DLBP and the effectiveness of the proposed MTLO algorithm, the personal computer case (McGovern & Gupta, 2007) containing eight disassembly tasks is introduced and modified. The relevant data information of the case, including the disassembly time (human worker/robot) of each task, the demand and hazardous indices of each component corresponding to each task, and the disassembly precedence relationships among the disassembly tasks, are shown in Table 2. The D and H in Table 2 indicate the demand and hazard indices. If the component corresponding to a disassembly task is remanufacturable,

Table 7

The parameters value settings for the algorithms.

Algorithm	The parameters values
MTLO	$Iter = 300, N = 200, rd = 5, Num_ls1 = 7, Lk = 10$
TLO	$Iter = 300, N = 200, Num_ls1 = 7, Lk = 10$
FASA	$Iter = 300, N = 200, x = 5, T_0 = 100, q = 0.9, Lk = 10$
NSGA-II	$Max_gen = 300, POP_num = 200, Pc = 0.92, Pm = 0.1$
GASA	$Gen = 300, T_0 = 100, T_{end} = 0.01, L = 10, q = 0.97$

the D indices of that task are set to 1; otherwise, 0. Similarly, if the component corresponds to a hazardous disassembly task, the indices H of that task is set to 1; otherwise, it is 0. In addition, the production cycle time of the personal computer HRC disassembly case is set to 40 s. The established MIP model is developed based on LINGO 11.0 software to obtain the optimal solution for each objective. At the same time, the proposed MTLO algorithm is applied to solve the case. The MTLO algorithm is implemented with MATLAB R2020a on a PC with the 11th Gen Intel(R) Core (TM) CPU (TM) i7-11800H @ 2.30 GHz and RAM 16.00 GB, and the values of the MTLO algorithm parameters are listed as follows: $Iter = 50, N = 50, rd = 3, Num_ls1 = 2, Lk = 3$.

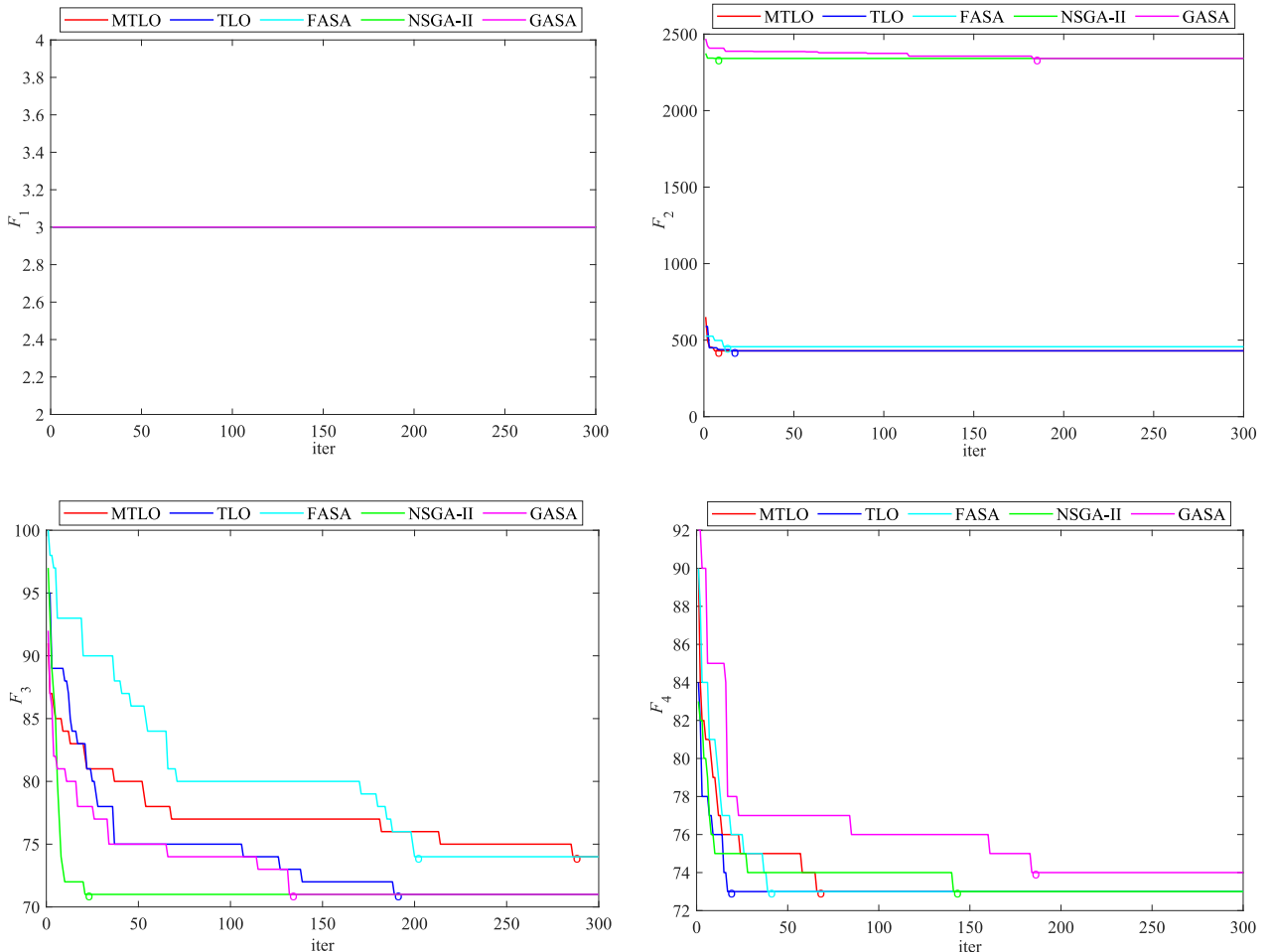
The results obtained by LINGO and the proposed MTLO algorithm are shown in Table 3, respectively. As shown in Table 3, the proposed MTLO algorithm obtained one solution for this personal computer case, and the solution results of the four optimization objectives correspond to the optimal solutions obtained by LINGO for each objective, respectively. The computation time of the proposed MTLO algorithm is 2.61 s, while it takes a longer time of 145 s for LINGO to obtain the optimal solution result for objective F_2 . Compared to the LINGO, the proposed

MTLO algorithm can acquire the optimal solutions for all four objectives in significantly less time. Thus, the correctness of the established MIP model and the effectiveness of the MTLO algorithm were preliminary verified.

Next, a theoretical case for a product consisting of 10 components (McGovern & Gupta, 2006) is modified, and the relevant data for the HRC disassembly case is shown in Table 4. The production cycle time is set to 40 s. The LINGO and the MTLO algorithm are applied to solve the case. The parameter values of the MTLO algorithm are listed as follows: $Iter = 100, N = 50, rd = 5, Num_ls1 = 3, Lk = 10$. Table 5 exhibits the solution results obtained by LINGO and the MTLO algorithm for the theoretical HRC disassembly case. A Pareto non-dominated solutions set, including 14 non-dominated solutions, is acquired by the MTLO algorithm. Therein, the non-dominated solutions 2, 4, 7 and 12 obtain the optimal value of 2 for objective F_1 ; the solutions 7 and 12 obtain the optimal value of 9 for objective F_2 ; the solutions 6, 8 and 11 obtain the optimal value of 3 for objective F_3 ; and the solutions 1 and 3 obtain the optimal value of 1 for objective F_4 . On the other hand, the proposed MTLO algorithm can obtain the optimal values of all objectives in 8.1 s, while it takes 947 s for LINGO to get the optimal value of objective F_2 . The above conclusion further validates the MIP model established for the HRC-DLBP and the effectiveness and efficiency of the proposed MTLO algorithm.

5.2. A medium-scale HRC disassembly case validation

This section uses a case of cathode ray tube (CRT) TV sets containing 27 disassembly tasks as a real-life case problem to validate the MTLO

**Fig. 7.** The iteration curve of each objective with five algorithms.

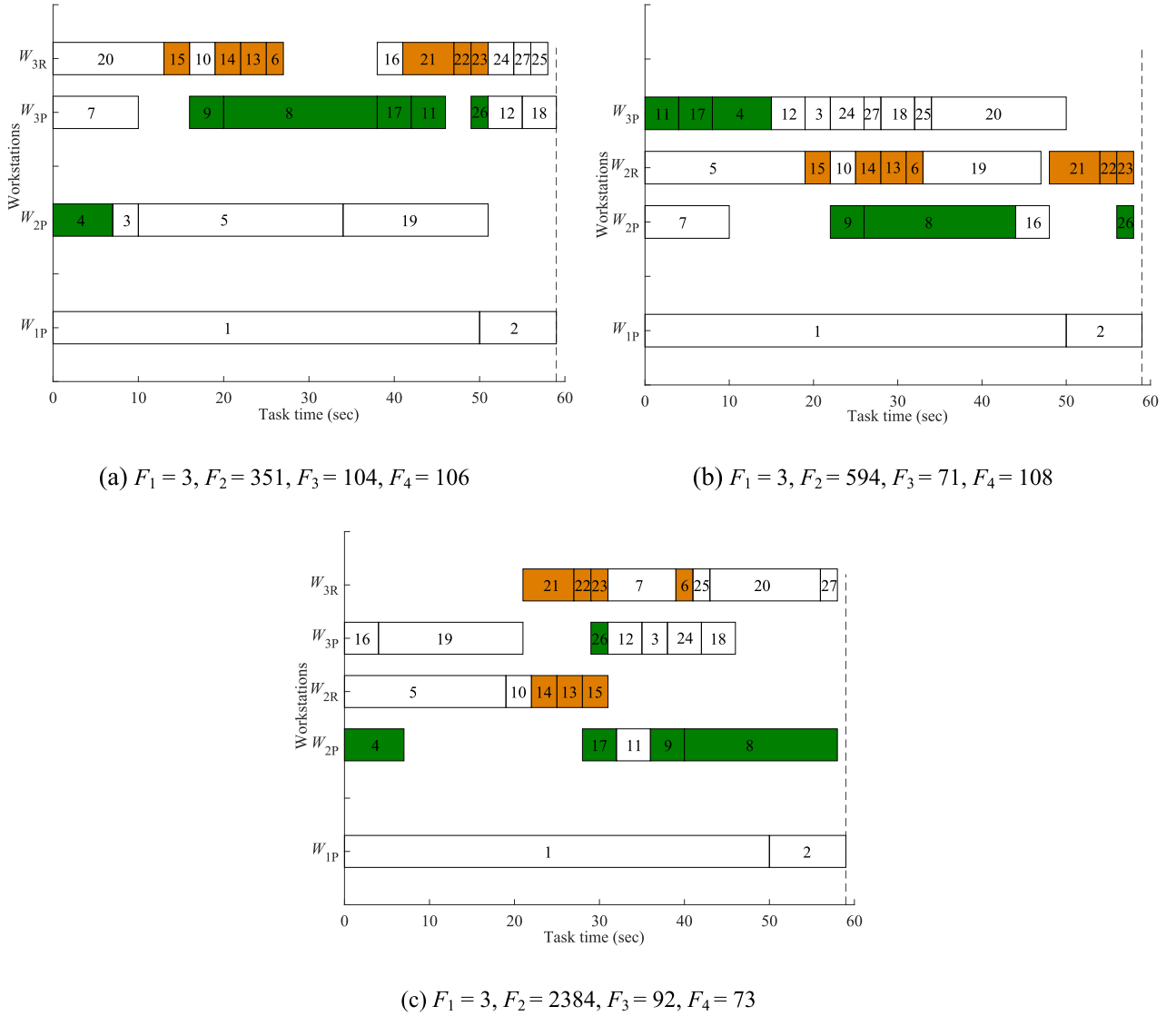


Fig. 8. Task allocation schemes corresponding to the optimal values of F_2 , F_3 , and F_4 three objectives obtained by the MTLO algorithm for CRT TV set HRC disassembly case.

algorithm. Moreover, the TLO, FASA (Zhang, Zhu, Chen, & Guan, 2023), GASA (Wang, Li, Gao, Li, & Gupta, 2021) and NSGA-II (Wu et al., 2022) algorithms are introduced for the comparative study with the proposed MTLO algorithm. The relevant data information required related to the case problem is given as follows: the production cycle time is 60 s, the disassembly time (human worker/robot) of each task, the demand indices, the hazard indices of each component corresponding to each task and the disassembly precedence relationships among the disassembly tasks can refer to Table 6. The parameter values settings of the proposed MTLO algorithm and four other comparison algorithms for the CRT TV HRC disassembly case, which are determined by a series of comprehensive comparative experiments are given in Table 7.

All the algorithms mentioned earlier were used to solve the case problem ten times to verify their statistical performance. Fig. 7 shows the iteration curves for each objective corresponding to the solution results obtained by the first calculation for the five algorithms. It can be seen from Fig. 7 that the FASA, NSGA-II and GASA converge to the minimum value of 457, 2341 and 2341, respectively, for objective F_2 , while the proposed MTLO and the TLO algorithms converge to the minimum value of 430 for objective F_2 . The convergence result of these two algorithms is better than the FASA, NSGA-II and GASA three algorithms. For objective F_3 , the MTLO, TLO, FASA, NSGA-II and GASA five

algorithms converge to the minimum value of 74, 71, 74, 71 and 71, respectively. The convergence result of the proposed MTLO algorithm on objective F_3 is inferior to the TLO, NSGA-II and GASA three algorithms. For objective F_4 , the MTLO, TLO, FASA, NSGA-II, and GASA algorithms converge to minimum values of 73, 73, 73, 73 and 74, respectively. The convergence result of the MTLO algorithm is equal to the TLO, FASA and NSGA-II three algorithms and is better than the GASA algorithm. When comparing the convergence results of these five algorithms across the four objectives, it is evident that, except for objective F_3 , where the convergence result of the proposed MTLO algorithm is inferior to the TLO, NSGA-II, and GASA algorithms, the MTLO algorithm demonstrates comparable or superior performance on the remaining three objectives.

Fig. 8 (a-c) illustrates the HRC disassembly task allocation scheme diagrams for the Pareto non-dominated solutions corresponding to the optimal values of F_2 , F_3 and F_4 obtained by the proposed MTLO algorithm. Since all Pareto non-dominated solutions obtained by the MTLO algorithm get the optimal value 3 for F_1 , the HRC disassembly task allocation schemes for the Pareto non-dominated solutions corresponding to the optimal values of F_2 , F_3 and F_4 are provided. Fig. 8 (a), (b) and (c) show the HRC disassembly task allocation scheme when achieving the minimum values obtained by MTLO as 351 for F_2 71 for F_3

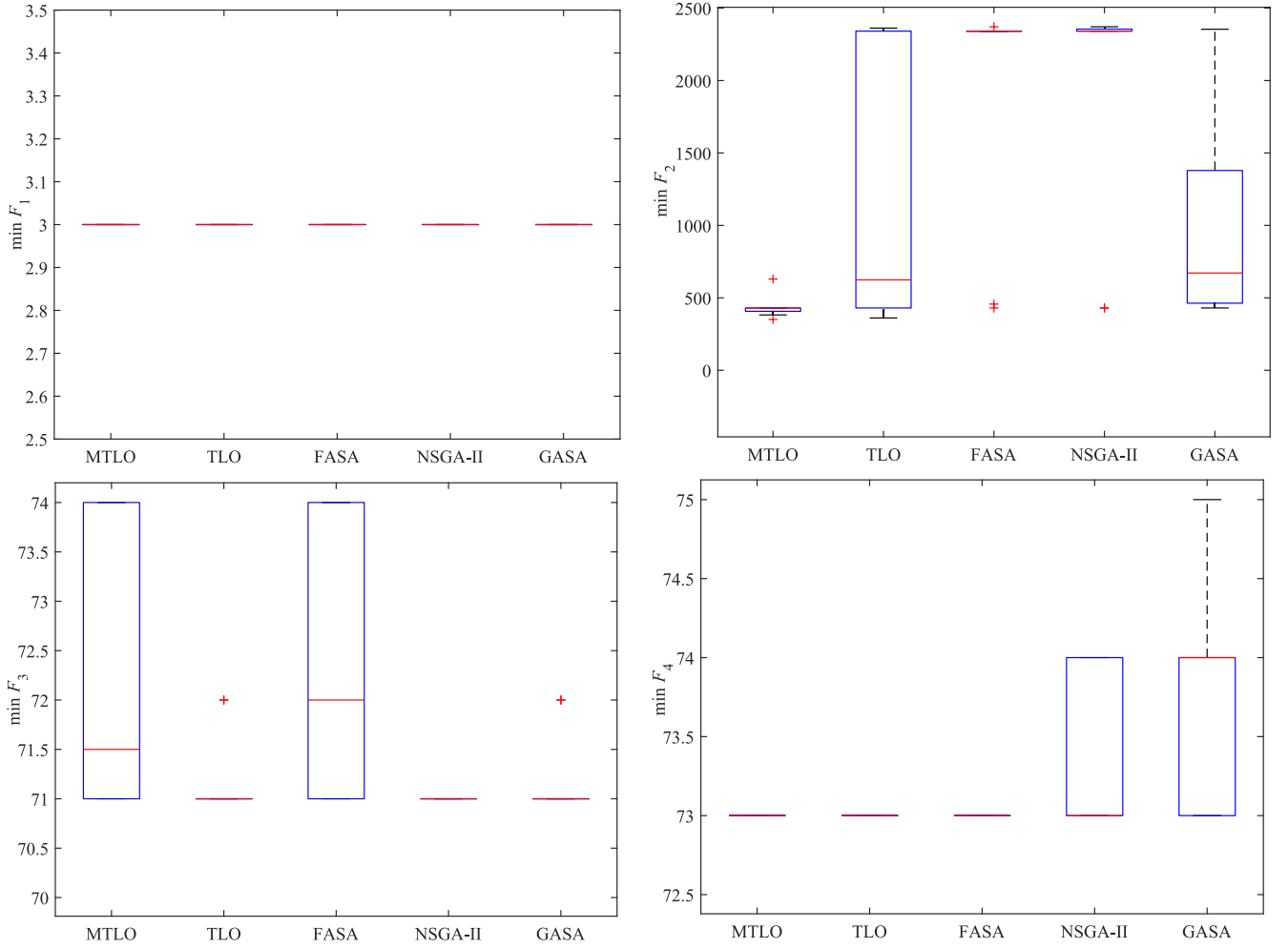


Fig. 9. Boxplots of minimum values of each objective of algorithms for CRT TV set HRC disassembly case.

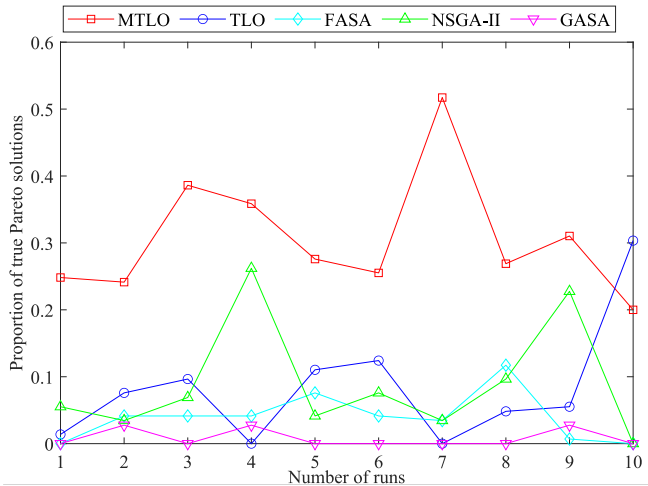


Fig. 10. The proportion of the number of the “true” Pareto optimal solutions of the Pareto non-dominated solutions obtained by the algorithms.

and 73 for F_4 , respectively. It can be noted that the orange box indicates the disassembly task corresponding to the hazardous component, and the green box indicates the disassembly task corresponding to the component with remanufacture demand (similarly in the following sections).

Fig. 9 exhibits the box plots of the minimum values obtained from ten calculations of five algorithms for the CRT TV set HRC disassembly case. It can be drawn from Fig. 9 that for the optimization objective F_1 , the solution results of the minimum value obtained by all algorithms are 3 in each calculation. For objective F_2 , the proposed MTLO algorithm exhibits narrower interquartile ranges and lower median values. However, the existence of outliers in the proposed MTLO algorithm's box plot represents the proposed algorithm's solving stability still needs to be improved. For objective F_3 , the NSGA-II algorithm can achieve the minimum value of 71 for each of the ten calculations, showing the best solving performance on objective F_3 . Both TLO and GASA two algorithms can obtain the minimum value of 71 in 8 out of 10 calculations. Although the proposed MTLO and the FASA two algorithms can obtain the minimum value 71 for objective F_3 , the overall solution results are worse than the other three. For objective F_4 , the proposed MTLO, the TLO and the FASA three algorithms can obtain the minimum value of 73 for each calculation. In comparison, the solving performance of the other two algorithms is not as good as the proposed MTLO, the TLO, and the FASA of these three algorithms.

Fig. 10 illustrates the statistics of the proportion of the “true” Pareto optimal solutions of the Pareto non-dominated solutions obtained by each algorithm. It is noted that the “true” Pareto optimal solutions are acquired by performing the filter operation on all Pareto non-dominated solutions obtained by ten calculations of all algorithms. Fig. 10 shows that the MTLO algorithm consistently yields a notably higher proportion of “true” Pareto optimal solutions than the other four algorithms across most calculations except for the 10th calculation, in which the TLO

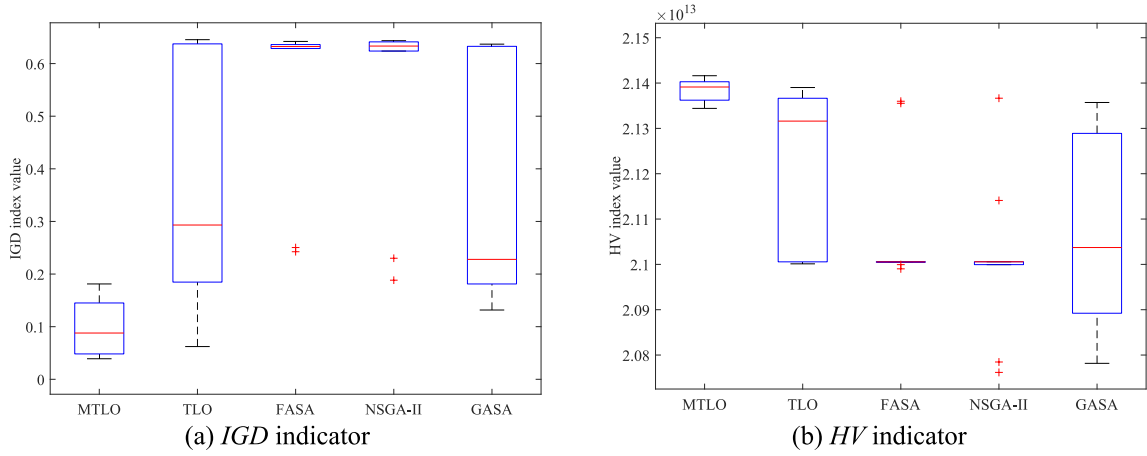


Fig. 11. Box plots of IGD and HV indicators of five algorithms for the CRT TV set HRC disassembly case.

Table 8

The related data for the printer HRC disassembly case.

Task	Time/s		D	H	Precedence disassembly tasks	Task	Time/s		D	H	Precedence disassembly tasks
	Human	Robot					Human	Robot			
1	3	2	0	0	—	29	15	12	0	0	25, 26, 27, 28
2	3	2	1	1	—	30	2	2	0	0	11
3	2	2	0	0	—	31	40	32	0	0	11
4	16	13	0	0	—	32	3	2	0	0	30, 31
5	5	4	0	0	4	33	8	6	0	0	11
6	40	32	0	0	—	34	2	2	1	0	33
7	18	14	0	0	11	35	24	19	0	0	11
8	8	6	0	0	—	36	2	2	1	1	35
9	33	26	0	0	8	37	15	12	0	0	32
10	8	6	0	0	—	38	2	2	0	0	32
11	13	10	0	0	10	39	32	26	0	0	7
12	5	4	0	0	—	40	36	29	0	0	39
13	16	13	0	0	—	41	32	26	0	0	—
14	10	8	0	0	9, 11, 12	42	4	3	0	0	27, 41
15	18	14	0	0	9	43	16	13	0	0	40
16	24	19	0	0	14	44	3	2	1	0	43, 48
17	4	3	1	0	16	45	8	6	0	0	11
18	8	6	0	0	—	46	2	2	0	0	45
19	2	2	0	0	18	47	8	6	0	0	40
20	16	13	0	0	9	48	2	2	0	0	40
21	32	26	0	0	9	49	2	2	0	0	40
22	15	12	1	1	15, 20, 21, 24, 25, 28	50	10	8	1	1	24, 30, 47, 48, 49
23	6	5	0	0	15	51	32	26	0	0	50
24	2	2	0	0	9	52	2	2	0	0	40
25	2	2	0	0	9	53	2	2	1	1	52
26	16	13	0	0	19	54	8	6	0	0	22
27	32	26	0	0	—	55	2	2	0	0	54
28	4	3	0	0	9						

Table 9

The algorithms parameter values settings for the printer HRC disassembly case.

Algorithm	The algorithm parameters values
MTLO	$Iter = 400, N = 300, rd = 15, Num_{ls1} = 5, Lk = 10$
TLO	$Iter = 400, N = 300, Num_{ls1} = 5, Lk = 10$
FASA	$Iter = 400, N = 150, x = 10, T_0 = 100, q = 0.92, Lk = 20$
NSGA-II	$Max_{gen} = 400, POP_{num} = 150, Pc = 0.92, Pm = 0.3$
GASA	$Gen = 400, T_0 = 100, T_{end} = 0.01, L = 30, q = 0.97$

algorithm obtains more “true” Pareto optimal solutions than the MTLO algorithm. However, after ten calculations, the total number of “true” Pareto optimal solutions obtained by the proposed MTLO algorithm is much more than the other four algorithms. Therefore, the superior multi-objective solving performance of the MTLO algorithm can be proved than the other four algorithms for the CRT TV set HRC disassembly case.

To comprehensively evaluate the multi-objective solving performance of the proposed MTLO algorithm, two multi-objective optimization evaluation metrics, i.e., inverted generational distance (IGD) and the hypervolume (HV), are employed to assess the multi-objective solving performance of the obtained Pareto non-dominated solutions set of each algorithm across convergence, distribution uniformity, and distribution universality several aspects. Fig. 11 (a-b) respectively represents the box plot of the IGD and HV indicators corresponding to the Pareto non-dominated solutions set obtained by ten calculations of the five algorithms. It can be seen from Fig. 11(a) that the IGD values of the Pareto non-dominated solutions sets obtained by ten calculations of the proposed MTLO, TLO, FASA, NSGA-II and GASA algorithms lie in the intervals (0.0391, 0.1812), (0.0622, 0.6454) (0.2425, 0.6421), (0.1884, 0.6435) and (0.1317, 0.6369) respectively. Fig. 11 (b) shows that the HV values of the Pareto non-dominated solutions sets obtained by ten calculations of the proposed MTLO, TLO, FASA, NSGA-II and GASA

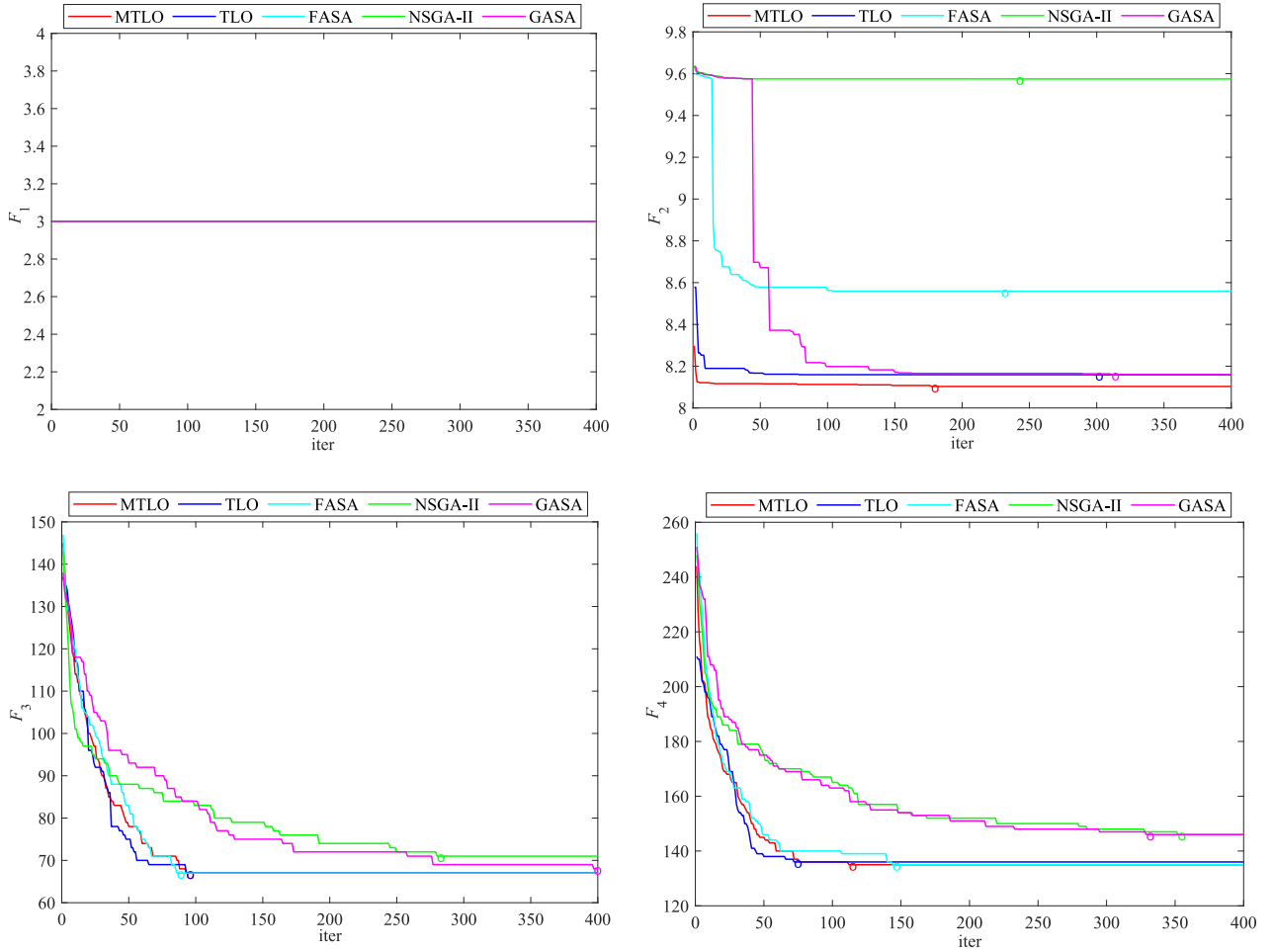


Fig. 12. The iteration curves of the four optimization objectives of the five algorithms for the printer HRC disassembly case.

algorithms lie in the intervals $(2.1344 \times 10^{13}, 2.1416 \times 10^{13})$, $(2.1001 \times 10^{13}, 2.1390 \times 10^{13})$, $(2.0990 \times 10^{13}, 2.1360 \times 10^{13})$, $(2.0762 \times 10^{13}, 2.1367 \times 10^{13})$ and $(2.0782 \times 10^{13}, 2.1357 \times 10^{13})$, respectively. The IGD values of the MTLO algorithm are generally smaller, and the HV values of the MTLO algorithm are generally more significant than the other four algorithms. In addition, the ranges of the interval values of the IGD and HV two indicators of the proposed MTLO algorithm are more concentrated than the other four algorithms. Therefore, based on the comparison results of the IGD and HV indicators, it can be concluded that the Pareto non-dominated solutions set obtained by the proposed MTLO algorithm are better than the TLO, FASA, NSGA-II and GASA algorithms in terms of the convergence, distribution uniformity and distribution universality three aspects.

5.3. A large-scale HRC disassembly case validation

A case problem for a specific type of printer containing 55 disassembly tasks is considered to verify the solving performance of the MTLO algorithm for large-scale problems against the compared algorithms. The cycle time of the printer case is 150 s. Other information data, including the disassembly precedence relationships among the tasks, the demand indices, the hazard indices of components corresponding to the tasks, and the disassembly time (human/robot) for each task, are given in Table 8. With the medium-scale case, the optimal values of the parameters for the five algorithms are determined through comprehensive comparative experiments shown in Table 9.

All algorithms are applied to solve the printer case problem ten times to validate the performance of the algorithms. Fig. 12 shows the

iteration curves of F_1 , F_2 , F_3 and F_4 of the solutions obtained by the first calculation of the five algorithms. As is shown in Fig. 12, for objective F_2 , the TLO, FASA, NSGA-II and GASA converge to the minimum value of 3495, 5211, 14,395 and 3498 in the 299th, 229th, 240th and 311th iterations, respectively. In contrast, the proposed MTLO algorithm converges to the minimum value of 3304 in the 177th generation. The convergence result of the MTLO algorithm is superior to those of the other algorithms. For objective F_3 , the MTLO, TLO and FASA converge to the minimum value of 67 in the 93rd, 93rd and 86th iterations, respectively. In comparison, the NSGA-II and the GASA converge to the minimum values of 71 and 68 in the 280th and 397th iterations. The convergence result of the proposed MTLO algorithm is equal to the TLO and FASA two algorithms and is better than the NSGA-II and GASA two algorithms. For objective F_4 , the MTLO and FASA two algorithms converge to the minimum value of 135, respectively in the 112th and 144th generations. Meanwhile, the TLO, NSGA-II, and GASA algorithms converge to the minimum values of 136, 146, and 146 in the 72nd, 352nd, and 329th generation. The convergence result of the proposed MTLO algorithm on objective F_4 is equal to the FASA and is superior to the other three algorithms. Thus, it can be concluded that the proposed MTLO algorithm can converge to the minimum value on all objectives. Notably, for objective F_2 , the MTLO algorithm outperforms the other four algorithms regarding convergence result and speed.

Fig. 13 (a-c) show the HRC disassembly task allocation schemes of the Pareto non-inferior solutions corresponding to the single-objective optimal solutions on F_2 , F_3 and F_4 obtained by the proposed MTLO algorithm for the printer HRC disassembly case, respectively. Since the objective function values of the objective F_1 of all Pareto non-dominated

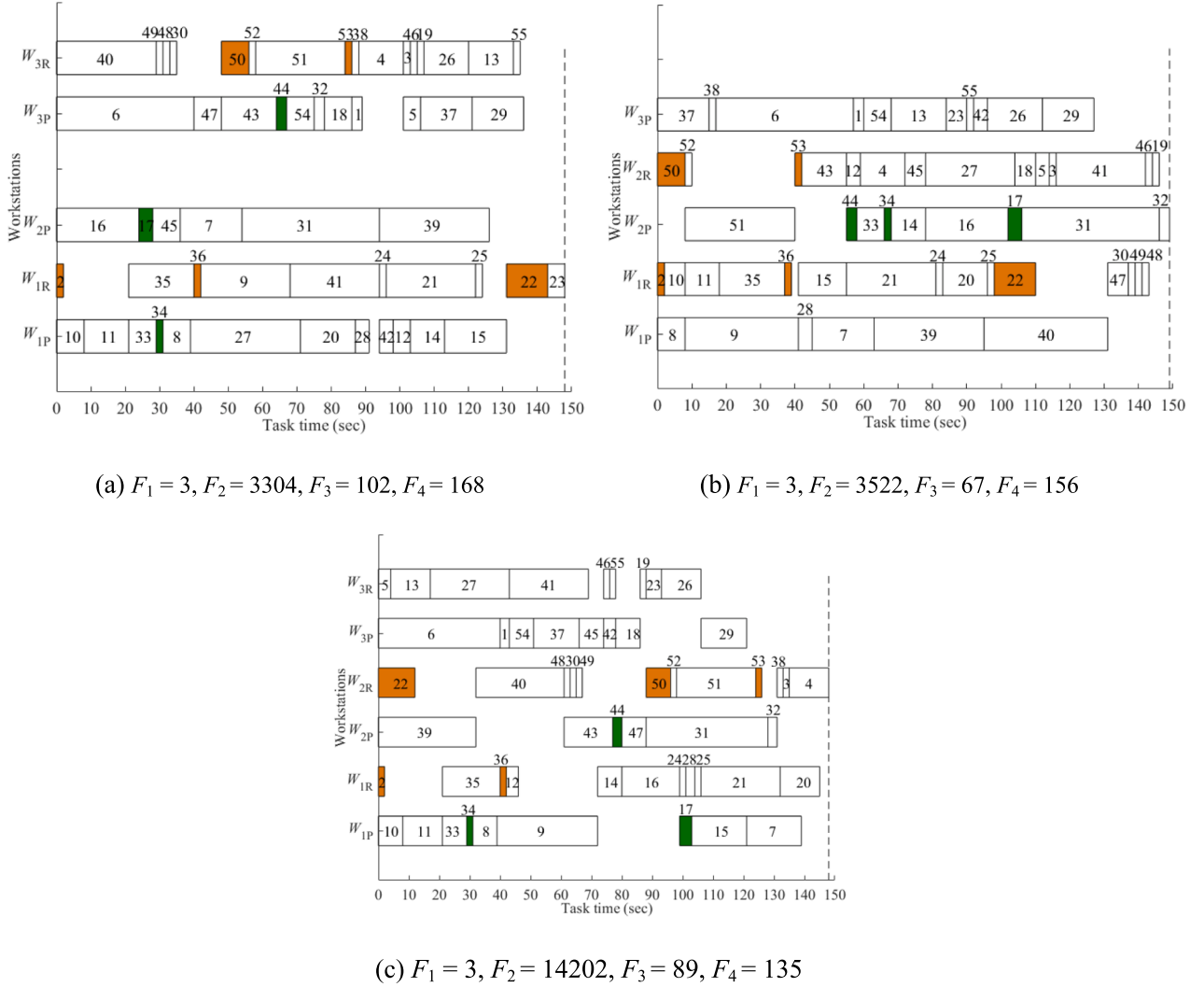


Fig. 13. The task allocation schemes corresponding to the single-objective optimal solutions obtained by the proposed MTLO algorithm for the printer HRC disassembly case.

solutions obtained by the MTLO algorithm is 3, the disassembly task allocation schemes for the Pareto non-dominated solutions corresponding to the optimal values of F_2 , F_3 and F_4 three objectives are provided. Fig. 13(a) shows the HRC disassembly task allocation scheme when achieving the minimum value of 3304 for objective F_2 . Fig. 13(b) shows the HRC disassembly task allocation scheme when achieving the minimum value 67 for objective F_3 . Fig. 13(c) shows the HRC disassembly task allocation scheme when achieving the minimum value 135 for objective F_4 .

Fig. 14 illustrates the boxplots of the minimum values of objectives for the ten runs of solution results obtained by the five algorithms. It can be observed that all non-dominated solutions obtain a minimum value of 3 for F_1 in each run. For F_2 , the values of the non-dominated solutions obtained by the proposed MTLO algorithm are better and more stable than the other algorithms. However, for F_3 and F_4 , the values of the non-dominated solutions of the proposed MTLO algorithm are close to those obtained by the TLO algorithm and are not as good as the FASA algorithm, though the solving performance of the MTLO algorithm on these two objectives are superior to the NSGA-II and GASA two algorithm. This observed superiority underscores the potential of MTLO as a proficient solution for addressing large-scale optimization challenges across diverse domains. Such empirical findings emphasize the significance of leveraging MTLO as a viable strategy for enhancing algorithmic

performance in real-world applications demanding scalability and efficacy in objective function minimization tasks.

Fig. 15 shows the statistical proportion of “true” Pareto optimal solutions of the Pareto non-dominated solutions obtained by the five algorithms. The TLO, NSGA-II and GASA algorithms cannot acquire the “true” Pareto optimal solutions in each run, and the FASA algorithm can only get one and three “true” Pareto optimal solutions, respectively in the 4th and 6th runs. In contrast, except for the 6th and 8th runs, the MTLO algorithm can obtain several “true” Pareto optimal solutions in each of the eight runs. However, the numbers of the “true” Pareto optimal solutions obtained by the eight runs of the MTLO algorithm are highly uneven, and only the non-dominated Pareto optimal solutions obtained by the 9th run accounted for more than 50 % of the “true” Pareto optimal solutions. On the one hand, it can be reflected from the side that the solution space of the printer case problem is relatively large and has more complexity. On the other hand, it can also be reflected from the side that the MTLO algorithm’s solving performance outperforms the other four algorithms.

Fig. 16 (a-b) shows the boxplot of the IGD and HV indices depicting the Pareto non-dominated solutions set obtained after ten algorithm runs. Fig. 16 (a) indicates that the IGD indices values for Pareto non-dominated solutions of the MTLO, TLO, FASA, NSGA-II and GASA algorithms lie in the intervals (0.0536, 0.2942), (0.1234, 0.8659),

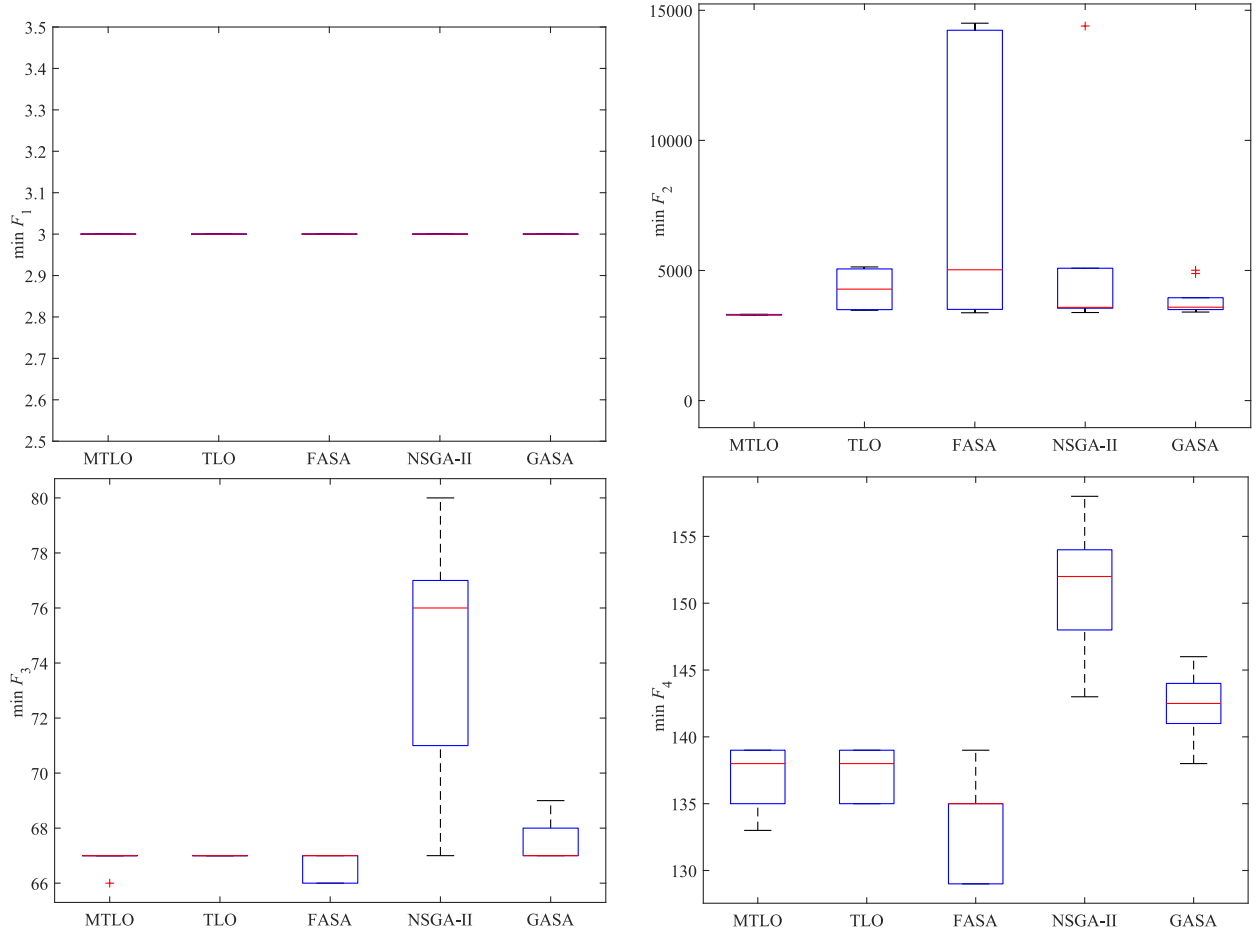


Fig. 14. Box plots of the minimum values of each objective obtained from 10 calculations using five algorithms for the printer HRC disassembly case.

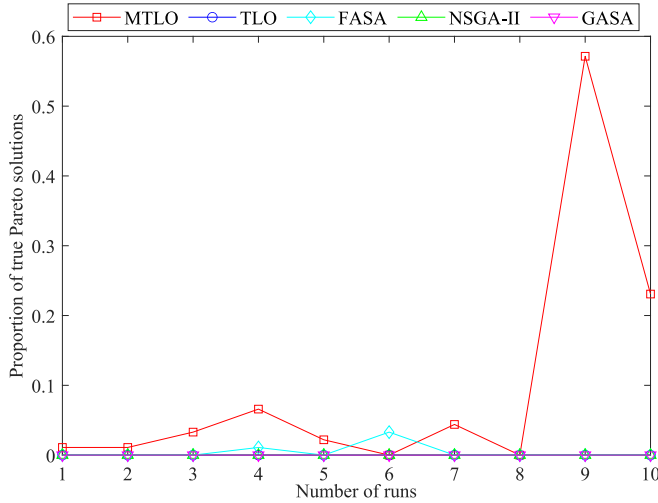


Fig. 15. The statistical proportion of "true" Pareto optimal solutions of Pareto non-dominated solutions.

(0.1971, 0.9023), (0.2937, 0.9043), (0.2085, 0.6095) respectively. The proposed MTLO algorithm exhibits narrower interquartile ranges and lower median values, proving that the non-dominated solutions obtained by the proposed MTLO algorithm are generally closer to the "true" Pareto optimal solutions. However, the existence of outliers in the box plot of the proposed MTLO algorithm represents the solving stability

of the proposed algorithm still needs to be improved. It can also be seen from Fig. 16 (b) that the HV indices values of the Pareto non-dominated solutions set of the MTLO, TLO, FASA, NSGA-II and GASA are in the intervals $(3.4628 \times 10^{12}, 3.5543 \times 10^{12})$, $(3.4520 \times 10^{12}, 3.5065 \times 10^{12})$, $(3.4293 \times 10^{12}, 3.5939 \times 10^{12})$, $(3.0720 \times 10^{12}, 3.4124 \times 10^{12})$ and $(3.3524 \times 10^{12}, 3.4676 \times 10^{12})$ respectively. The performance of the MTLO, TLO and FASA three algorithms on the HV indicator are similar and better than the NSGA-II and GASA algorithms. However, the interval range of the FASA algorithm is larger than the MTLO and TLO two algorithms. The median value is slightly higher than the MTLO and TLO, indicating that the overall HV indicator performance is superior to the MTLO and TLO algorithms. The proposed MTLO algorithm's performance in solving objectives F_3 and F_4 is insufficient, resulting in limited coverage of the Pareto non-dominated solutions set in the target space. The overall comparison shows that the Pareto non-dominated solutions sets obtained by the MTLO algorithm are better than those of the other four algorithms regarding convergence and distribution uniformity, and there is still room for improvement in solving diversity and stability issues.

6. Conclusion

In this study, we addressed the critical problem of human-robot collaborative disassembly line balancing, in which the disassembly tasks are classified into demand, hazard, and general tasks based on the remanufacturable demand and hazard attributes of the components corresponding to each task and a mapping constraint mechanism between the disassembly tasks and the task operators is established. We developed a comprehensive mixed-integer programming model

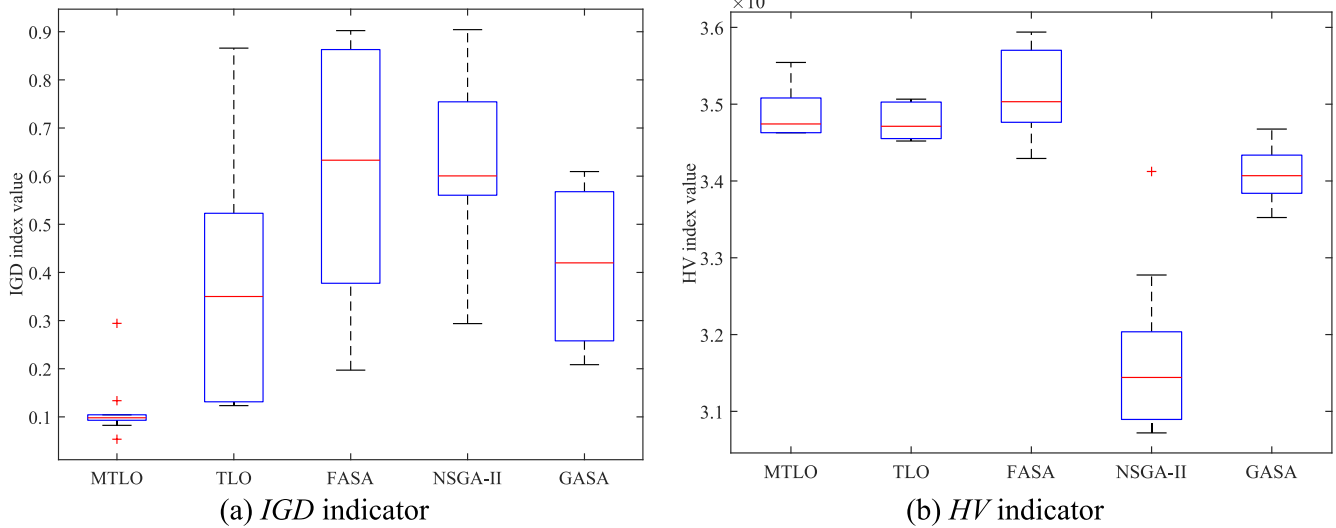


Fig. 16. Box plots of IGD and HV indicators of five algorithms for the printer HRC disassembly case.

incorporating four key optimization objectives: the number of workstations, the task operator idle balancing index, the demand index, and the hazard index. To solve this complex problem, a novel modified teaching and learning optimization, i.e., the MTLO approach, is proposed. The two-point crossover operation is introduced into the teaching stage, and the single-point insertion and the two-point swap operation are applied in the learning stage. Besides, the perturbation strategy based on randomly transforming the task operator is introduced in the teaching and learning stages to enhance the optimization performance of the MTLO algorithm. The correctness of the established mixed-integer programming model and the effectiveness of the presented MTLO algorithm are verified through two small-scale HRC disassembly cases. Then, two HRC disassembly cases of different scales modified from the existing literature are introduced. The proposed MTLO algorithm and the other four advanced algorithms are applied to solve the case problems. The solution results of the five algorithms are compared and analyzed from the aspects of single-objective and multi-objective optimization performance, including convergence, distribution uniformity and distribution universality. The comparison results show that for the single-objective optimization performance, the MTLO algorithm has good solving performance for objective F_2 , but its exploration ability for objectives F_3 and F_4 is still insufficient. In addition, the proposed MTLO algorithm can obtain more “true” Pareto non-dominated solutions than the other four algorithms, which proves the excellent multi-objective optimization performance; however, the solving diversity of the proposed MTLO algorithm needs to be improved.

The limitations of this study are as follows: firstly, the human-robot collaborative disassembly line balancing problem addressed in this paper is limited to human-robot collaboration, that is, human workers and robots collaborate independently to complete disassembly production, the human-robot interaction operation is not involved. Besides, the problem research is based on a deterministic disassembly production environment and many uncertain factors are ignored. In addition, there is still room for further improvement of the diversity and stability of the proposed algorithm in solving similar human-machine collaborative disassembly line balancing problems. Future research could further refine the model by incorporating more dynamic factors and exploring additional optimization techniques to continue improving the performance and applicability of human-robot collaborative disassembly line balancing solution schemes.

CRediT authorship contribution statement

Lixia Zhu: Conceptualization, Writing – original draft, Resources. **Yarong Chen:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Jabir Mumtaz:** Writing – review & editing, Visualization, Supervision, Resources, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work has been supported by the Basic Scientific Research Project of Wenzhou City (G20210024), (G2023036) and (G20240020).

References

- Baratta, A., Cimino, A., Longo, F., & Nicoletti, L. (2023). Digital twin for human-robot collaboration enhancement in manufacturing systems: Literature review and direction for future developments. *Computers & Industrial Engineering*, 109764.
- Bentaha, M. L., Dolgui, A., Battaia, O., Riggs, R. J., & Hu, J. (2018). Profit-oriented partial disassembly line design: Dealing with hazardous parts and task processing times uncertainty. *International Journal of Production Research*, 56(24), 7220–7242.
- Çil, Z. A., Kizilay, D., Li, Z., & Öztup, H. (2022). Two-sided disassembly line balancing problem with sequence-dependent setup time: A constraint programming model and artificial bee colony algorithm. *Expert Systems with Applications*, 203, Article 117529.
- Du, S., Xu, R., & Li, L. (2018). Modeling and analysis of multiproduct multistage manufacturing system for quality improvement. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 48(5), 801–820. <https://doi.org/10.1109/TSMC.2016.2614766>
- Fang, Y., Li, Z., Wang, S., & Lu, X. (2024). Multi-objective multi-fidelity optimisation for position-constrained human-robot collaborative disassembly planning. *International Journal of Production Research*, 62(11), 3872–3889.
- Fang, Y., Ming, H., Li, M., Liu, Q., & Pham, D. T. (2020). Multi-objective evolutionary simulated annealing optimisation for mixed-model multi-robotic disassembly line balancing with interval processing time. *International Journal of Production Research*, 58(3), 846–862.
- Guo, H., Zhang, L., Ren, Y., Li, Y., Zhou, Z., & Wu, J. (2023). Optimizing a stochastic disassembly line balancing problem with task failure via a hybrid variable neighborhood descent-artificial bee colony algorithm. *International Journal of Production Research*, 61(7), 2307–2321.

- Guo, J., Li, Y., Du, B., Sun, X., & Wang, K. (2024). A multi-population cooperative coevolution artificial bee colony algorithm for partial multi-robotic disassembly line balancing problem considering preventive maintenance scenarios. *Advanced Engineering Informatics*, 62, Article 102750.
- Guo, L., Zhang, Z., Wu, T., Zhang, Y., Zeng, Y., & Xie, X. (2024). Green and efficient-oriented human-robot hybrid partial destructive disassembly line balancing problem from non-disassemblability of components and noise pollution. *Robotics and Computer-Integrated Manufacturing*, 90, Article 102816.
- Guo, X., Fan, C., Zhou, M., Liu, S., Wang, J., Qin, S., & Tang, Y. (2023). Human-robot collaborative disassembly line balancing problem with stochastic operation time and a solution via multi-objective shuffled frog leaping algorithm. *IEEE Transactions on Automation Science and Engineering*.
- Guo, X., Wei, T., Wang, J., Liu, S., Qin, S., & Qi, L. (2022). Multiobjective U-shaped disassembly line balancing problem considering human fatigue index and an efficient solution. *IEEE Transactions on Computational Social Systems*.
- Hartono, N., Ramirez, F. J., & Pham, D. T. (2023). A multiobjective decision-making approach for modelling and planning economically and environmentally sustainable robotic disassembly for remanufacturing. *Computers & Industrial Engineering*, 184, Article 109535.
- Hjorth, S., & Chrysostomou, D. (2022). Human-robot collaboration in industrial environments: A literature review on non-destructive disassembly. *Robotics and Computer-Integrated Manufacturing*, 73, Article 102208.
- Hu, Y., Liu, C., Zhang, M., Lu, Y., Jia, Y., & Xu, Y. (2024). An ontology and rule-based method for human-robot collaborative disassembly planning in smart remanufacturing. *Robotics and Computer-Integrated Manufacturing*, 89, Article 102766.
- Jiao, J., Feng, G., & Yuan, G. (2024). Research on the human-robot collaborative disassembly line balancing of spent lithium batteries with a human factor load. *Batteries*, 10(6), 196.
- Jin, L., Zhang, X., Fang, Y., & Pham, D. T. (2022). Transfer learning-assisted evolutionary dynamic optimisation for dynamic human-robot collaborative disassembly line balancing. *Applied Sciences*, 12(21), 11008.
- Kheirabadi, M., Keivanpour, S., Chinniah, Y. A., & Frayret, J.-M. (2023). Human-robot collaboration in assembly line balancing problems: Review and research gaps. *Computers & Industrial Engineering*, 186, Article 109737.
- Lee, M.-L., Behdad, S., Liang, X., & Zheng, M. (2022). Task allocation and planning for product disassembly with human-robot collaboration. *Robotics and Computer-Integrated Manufacturing*, 76, Article 102306.
- Liang, W., Zhang, Z., Zhang, Y., Xu, P., & Yin, T. (2023). Improved social spider algorithm for partial disassembly line balancing problem considering the energy consumption involved in tool switching. *International Journal of Production Research*, 61(7), 2250–2266.
- Liu, J., Zhou, Z., Pham, D. T., Xu, W., Ji, C., & Liu, Q. (2020). Collaborative optimization of robotic disassembly sequence planning and robotic disassembly line balancing problem using improved discrete Bees algorithm in remanufacturing. *Robotics and Computer-Integrated Manufacturing*, 61, Article 101829.
- Liu, M., Lv, J., Du, S., Deng, Y., Shen, X., & Zhou, Y. (2024). Multi-resource constrained flexible job shop scheduling problem with fixture-pallet combinatorial optimisation. *Computers & Industrial Engineering*, 188, Article 109903.
- Lou, S., Tan, R., Zhang, Y., Zhou, M., & Lv, C. (2024). Personalized disassembly sequence planning for a human-robot hybrid disassembly cell. *IEEE Transactions on Industrial Informatics*.
- McGovern, S. M., & Gupta, S. M. (2006). Ant colony optimization for disassembly sequencing with multiple objectives. *The International Journal of Advanced Manufacturing Technology*, 30, 481–496.
- McGovern, S. M., & Gupta, S. M. (2007). A balancing method and genetic algorithm for disassembly line balancing. *European Journal of Operational Research*, 179(3), 692–708.
- Mumtaz, J., Guan, Z., Yue, L., Wang, Z., Ullah, S., & Rauf, M. (2019). Multi-level planning and scheduling for parallel PCB assembly lines using hybrid spider monkey optimization approach. *Ieee Access*, 7, 18685–18700.
- Mumtaz, J., Guan, Z., Yue, L., Zhang, L., & He, C. (2020). Hybrid spider monkey optimization algorithm for multi-level planning and scheduling problems of assembly lines. *International Journal of Production Research*, 58(20), 6252–6267.
- Mumtaz, J., Minhas, K. A., Rauf, M., Yue, L., & Chen, Y. (2024). Solving line balancing and AGV scheduling problems for intelligent decisions using a Genetic-Artificial bee colony algorithm. *Computers & Industrial Engineering*, 189, Article 109976.
- Mumtaz, J., Zailin, G., Mirza, J., Rauf, M., Sarfraz, S., & Shehab, E. (2018). Makespan minimization for flow shop scheduling problems using modified operators in genetic algorithm. In *Advances in Manufacturing Technology XXXII* (pp. 435–440). IOS Press.
- Parsa, S., & Saadat, M. (2021). Human-robot collaboration disassembly planning for end-of-life product disassembly process. *Robotics and Computer-Integrated Manufacturing*, 71, Article 102170.
- Qin, S., Li, C., Wang, J., Liu, S., Zhao, Z., Guo, X., & Qi, L. (2024). Multiobjective human-robot collaborative disassembly sequence planning: considering the properties of components. *IEEE Systems, Man, and Cybernetics Magazine*, 10(2), 15–23.
- Qin, S., Wang, J., Wang, J., Liu, S., Guo, X., Qi, L., & Zhao, Z. (2024). An improved fruit fly optimization algorithm for disassembly lines requiring multiskilled workers. *IEEE Transactions on Computational Social Systems*.
- Qin, S., Xiang, C., Wang, J., Liu, S., Guo, X., & Qi, L. (2024). Multiple product hybrid disassembly line balancing problem with human-robot collaboration. *IEEE Transactions on Automation Science and Engineering*.
- Rinaldi, M., Caterino, M., & Fera, M. (2023). Sustainability of Human-Robot cooperative configurations: Findings from a case study. *Computers & Industrial Engineering*, 182, Article 109383.
- Wang, G., Chen, Y., Mumtaz, J., & Zhu, L. (2023). A study of mixed-flow human-machine collaborative disassembly line balancing problem based on improved artificial fish swarm algorithm. *Engineering Proceedings*, 45(1), 40.
- Wang, K., Guo, J., Du, B., Li, Y., Tang, H., Li, X., & Gao, L. (2023). A novel MILP model and an improved genetic algorithm for disassembly line balancing and sequence planning with partial destructive mode. *Computers & Industrial Engineering*, 186, Article 109704.
- Wang, K., Li, X., Gao, L., Li, P., & Gupta, S. M. (2021). A genetic simulated annealing algorithm for parallel partial disassembly line balancing problem. *Applied Soft Computing*, 107, Article 107404.
- Wu, T., Zhang, Z., Yin, T., & Zhang, Y. (2022). Multi-objective optimisation for cell-level disassembly of waste power battery modules in human-machine hybrid mode. *Waste Management*, 144, 513–526.
- Wu, T., Zhang, Z., Zeng, Y., & Zhang, Y. (2024). Mixed-integer programming model and hybrid local search genetic algorithm for human-robot collaborative disassembly line balancing problem. *International Journal of Production Research*, 62(5), 1758–1782.
- Wu, T., Zhang, Z., Zeng, Y., Zhang, Y., Guo, L., & Liu, J. (2024). Techno-economic and environmental benefits-oriented human-robot collaborative disassembly line balancing optimization in remanufacturing. *Robotics and Computer-Integrated Manufacturing*, 86, Article 102650.
- Wu, T., Zhang, Z., Zhang, Y., & Zeng, Y. (2023). Modelling and optimisation of two-sided disassembly line balancing problem with human-robot interaction constraints. *Expert Systems with Applications*, 230, Article 120589.
- Xu, W., Cui, J., Liu, B., Liu, J., Yao, B., & Zhou, Z. (2021). Human-robot collaborative disassembly line balancing considering the safe strategy in remanufacturing. *Journal of Cleaner Production*, 324, Article 129158.
- Xu, W., Tang, Q., Liu, J., Liu, Z., Zhou, Z., & Pham, D. T. (2020). Disassembly sequence planning using discrete Bees algorithm for human-robot collaboration in remanufacturing. *Robotics and Computer-Integrated Manufacturing*, 62, Article 101860.
- Yao, B., Li, X., Ji, Z., Xiao, K., & Xu, W. (2024). Task reallocation of human-robot collaborative production workshop based on a dynamic human fatigue model. *Computers & Industrial Engineering*, 189, Article 109855.
- Yin, T., Zhang, Z., Liang, W., Zeng, Y., & Zhang, Y. (2023). Multi-Man-Robot Disassembly Line Balancing Optimization by Mixed-Integer Programming and Problem-Oriented Group Evolutionary Algorithm. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*.
- Yin, T., Zhang, Z., Wu, T., Zeng, Y., Zhang, Y., & Liu, J. (2023). Multimanned partial disassembly line balancing optimization considering end-of-life states of products and skill differences of workers. *Journal of Manufacturing Systems*, 66, 107–126.
- Yin, T., Zhang, Z., Zhang, Y., Wu, T., & Liang, W. (2022). Mixed-integer programming model and hybrid driving algorithm for multi-product partial disassembly line balancing problem with multi-robot workstations. *Robotics and Computer-Integrated Manufacturing*, 73, Article 102251.
- Zeng, Y., Zhang, Z., Liang, W., & Zhang, Y. (2023). Balancing optimization for disassembly line of mixed homogeneous products with hybrid disassembly mode. *Computers & Industrial Engineering*, 185, Article 109646.
- Zeng, Y., Zhang, Z., Yin, T., & Zheng, H. (2022). Robotic disassembly line balancing and sequencing problem considering energy-saving and high-profit for waste household appliances. *Journal of Cleaner Production*, 381, Article 135209.
- Zhang, Q., Xu, B., Yao, M., Wang, J., Guo, X., Qin, S., & Lu, F. (2024). An improved moth-flame algorithm for human-robot collaborative parallel disassembly line balancing problem. *Mathematics*, 12(6), 816.
- Zhang, Z., Zhu, L., Chen, Y., & Guan, C. (2023). A multi-objective hybrid evolutionary search algorithm for parallel production line balancing problem including disassembly and assembly tasks. *International Transactions in Operational Research*, 30 (6), 3508–3553.
- Zhou, B., & Bian, J. (2022a). A bi-objective salp swarm algorithm with sine cosine operator for resource constrained multi-manned disassembly line balancing problem. *Applied Soft Computing*, 131, Article 109759.
- Zhou, B., & Bian, J. (2022b). Multi-mechanism-based modified bi-objective Harris Hawks optimization for sustainable robotic disassembly line balancing problems. *Engineering Applications of Artificial Intelligence*, 116, Article 105479.
- Zhou, Y., Du, S., Liu, M., & Shen, X. (2024). Machine-fixture-pallet resources constrained flexible job shop scheduling considering loading and unloading times under pallet automation system. *Journal of Manufacturing Systems*, 73, 143–158.
- Zhu, L., Zhang, Z., & Guan, C. (2020). Multi-objective partial parallel disassembly line balancing problem using hybrid group neighbourhood search algorithm. *Journal of Manufacturing Systems*, 56, 252–269.